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Data Warehouse Using Snowflake

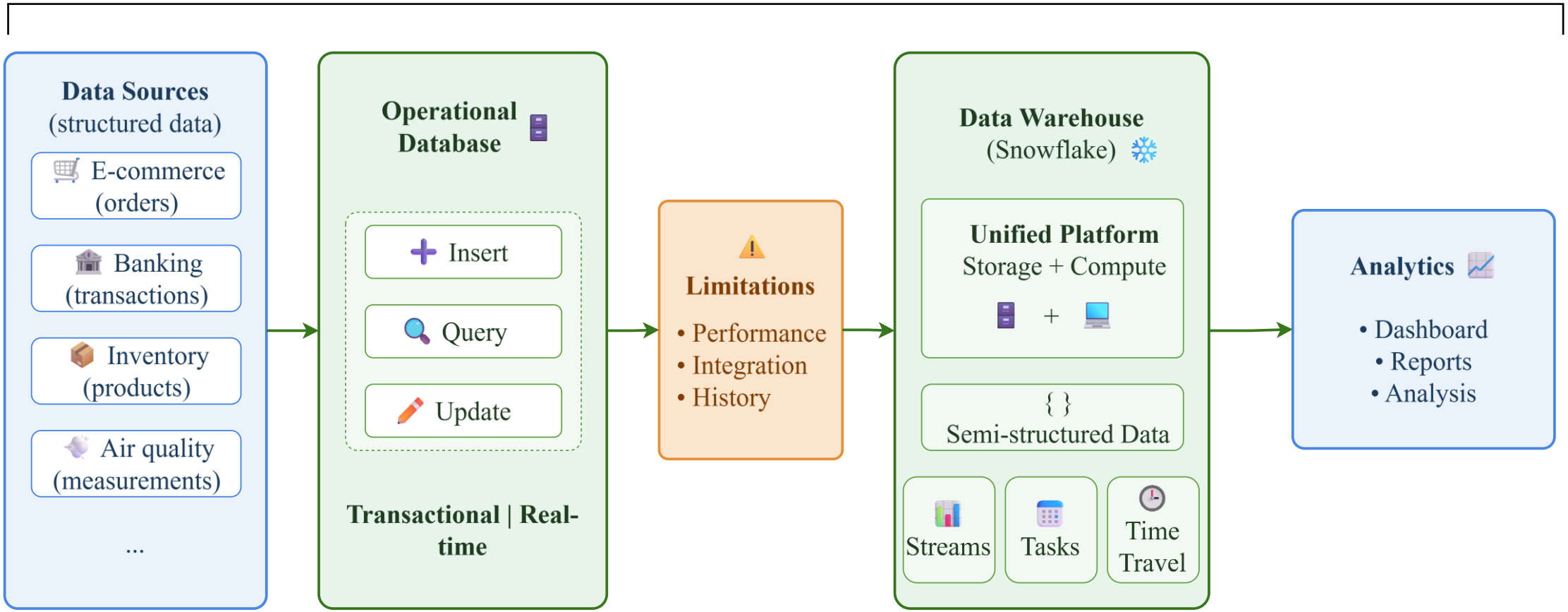
Nguyen Dang Nha - Senior TA
Tran Quang Minh - TA

Proofreaders: Phạm Tiến – QA/QC: Nguyễn Thông

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Motivation

Why we need Data Warehouse and Snowflake?

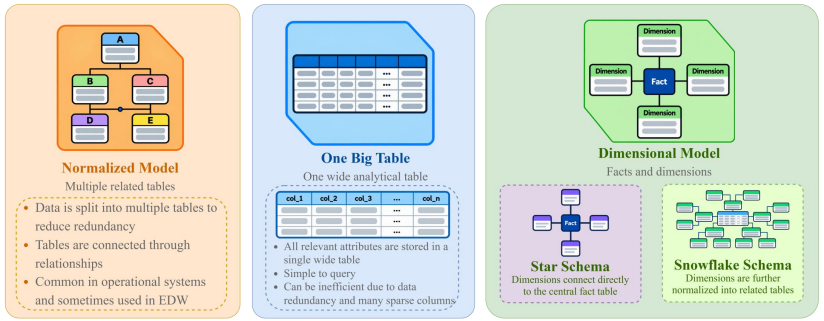


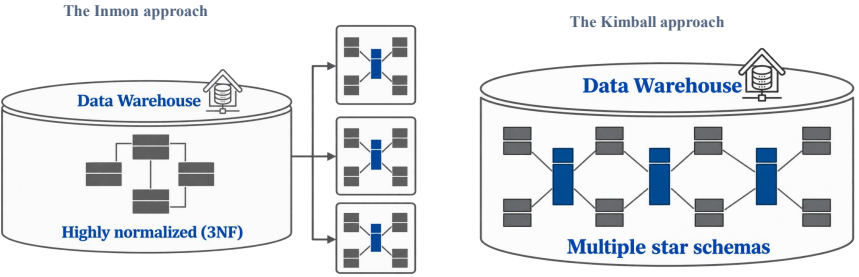
The flow of Data go from Sources to Analystics target

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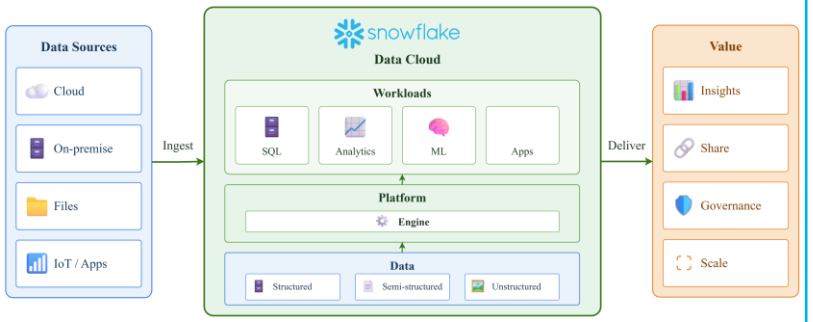
Objective

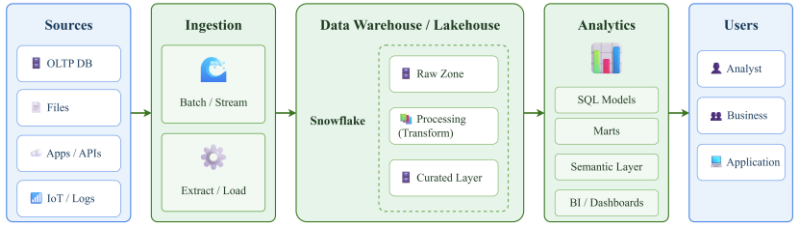
Data Warehouse



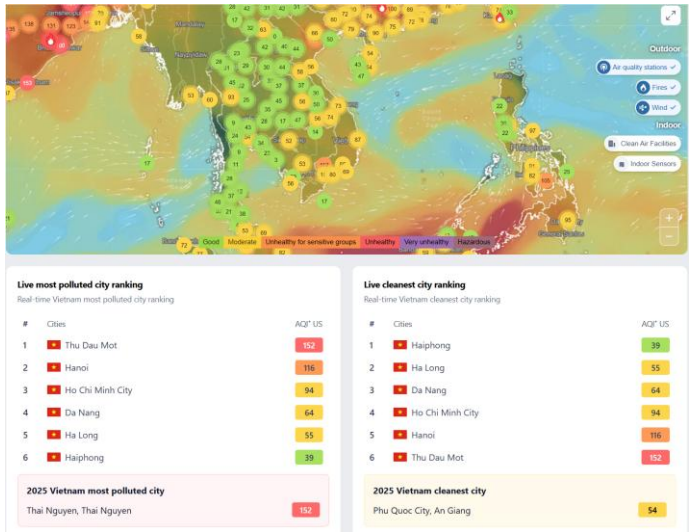


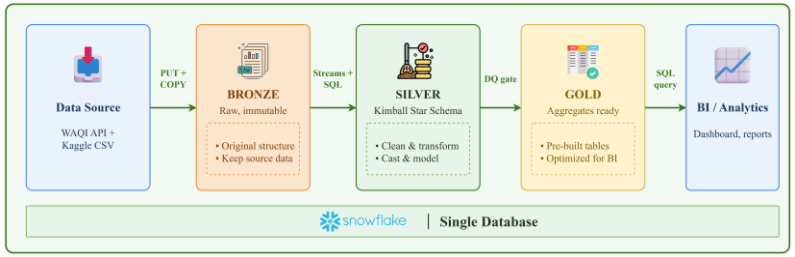
Snowflake





Vietnam Air Quality





Outline

- Introduction
- From Database to Data Warehouse
- Snowflake
- Project: Vietnam Air Quality Data Pipeline
- Q&A

Quiz&Feedbacks

Introduction

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Introduction

❖ Where is data come from?



Customer

Buy

CIRCLE K

VINCOM

WinMart

7 ELEVEN

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100

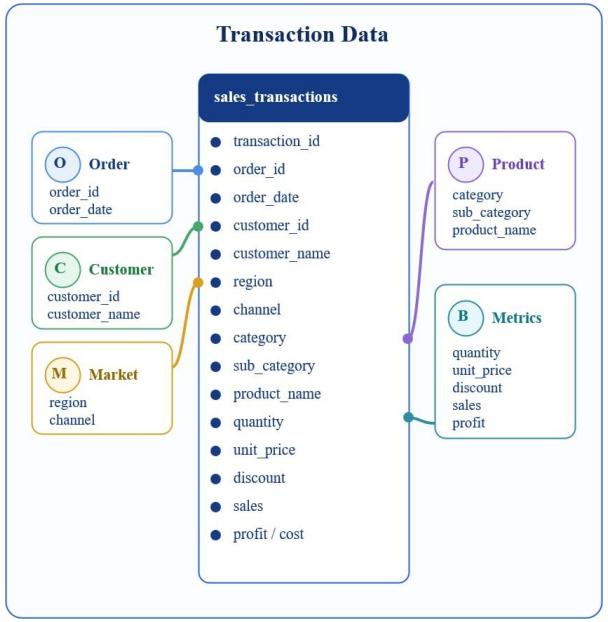


Customer

Buy

TikTok Shop

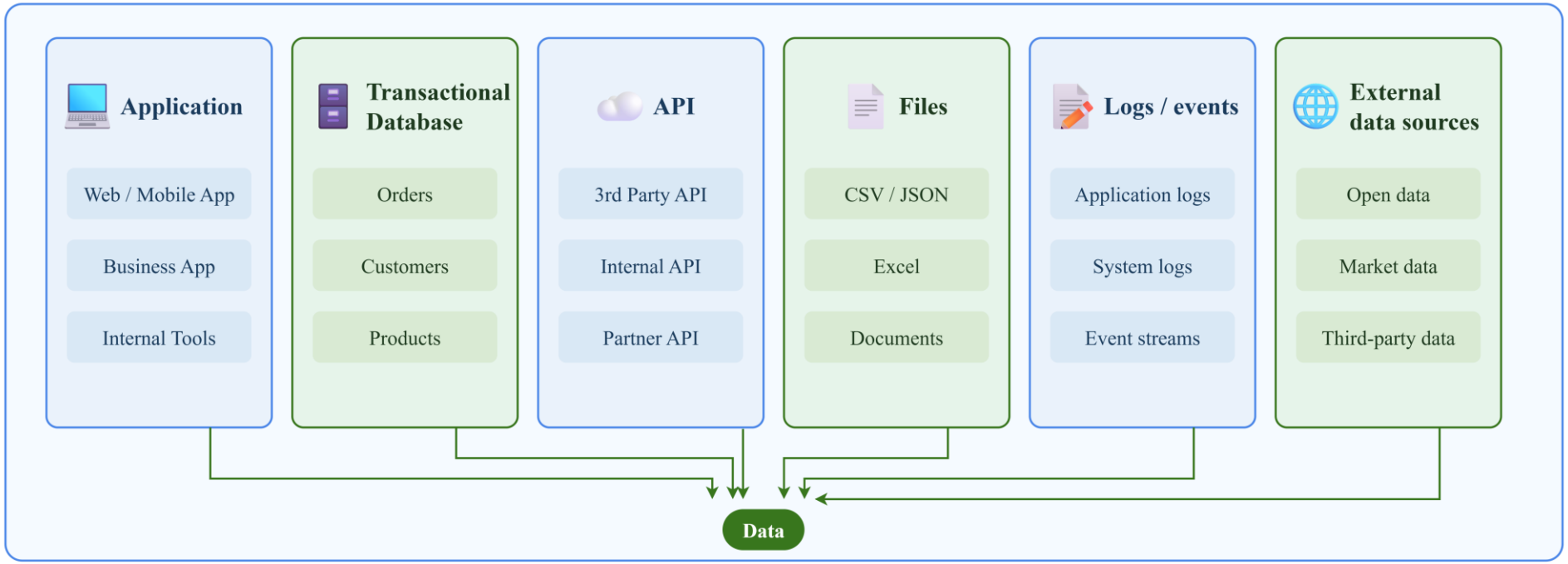
Shopee

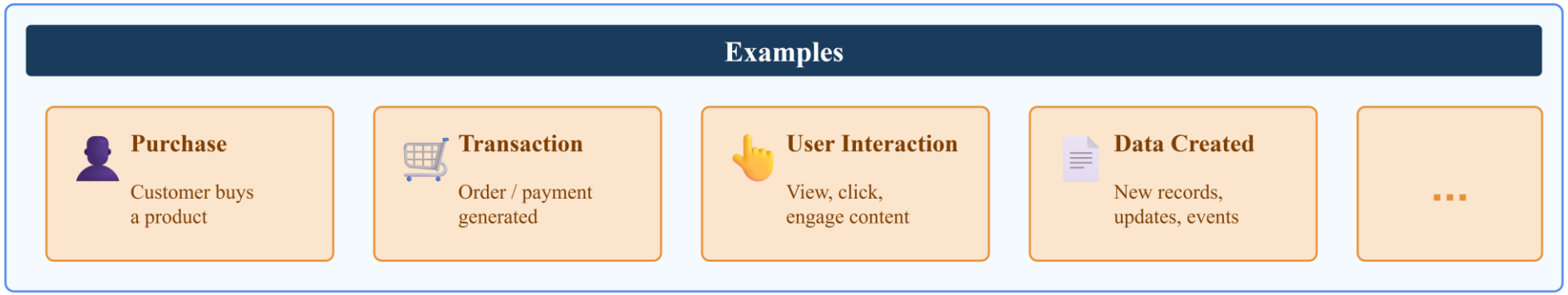


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Introduction

❖ Where is data come from?





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Introduction

❖ Data structure



Organize

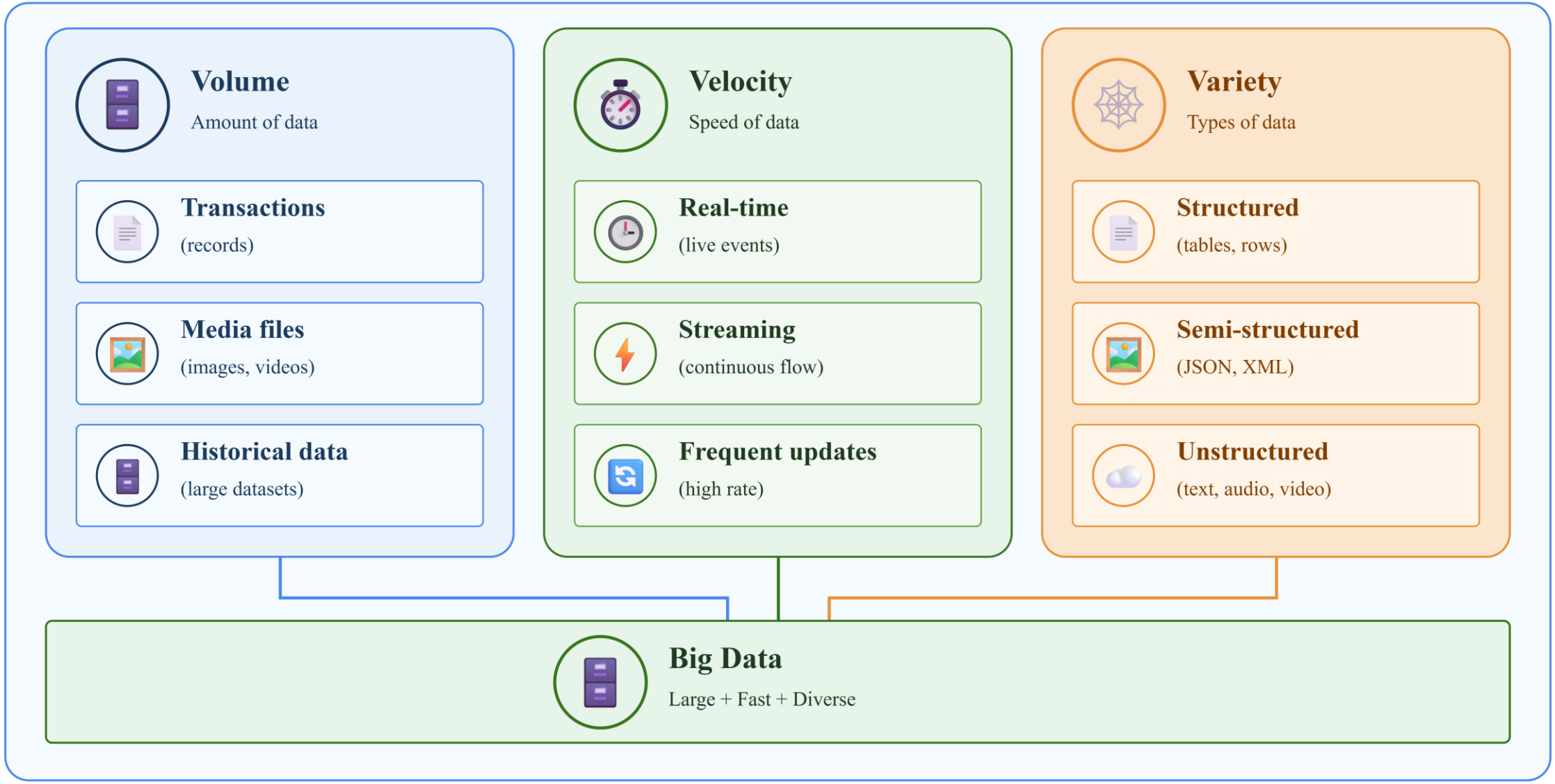


Unstructured	Semi-Structured	Structured
PDF, MP3, JPEG, ...	CSV, mongoDB, JSON, XML, ...	ORACLE, MySQL, Microsoft SQL Server, ...

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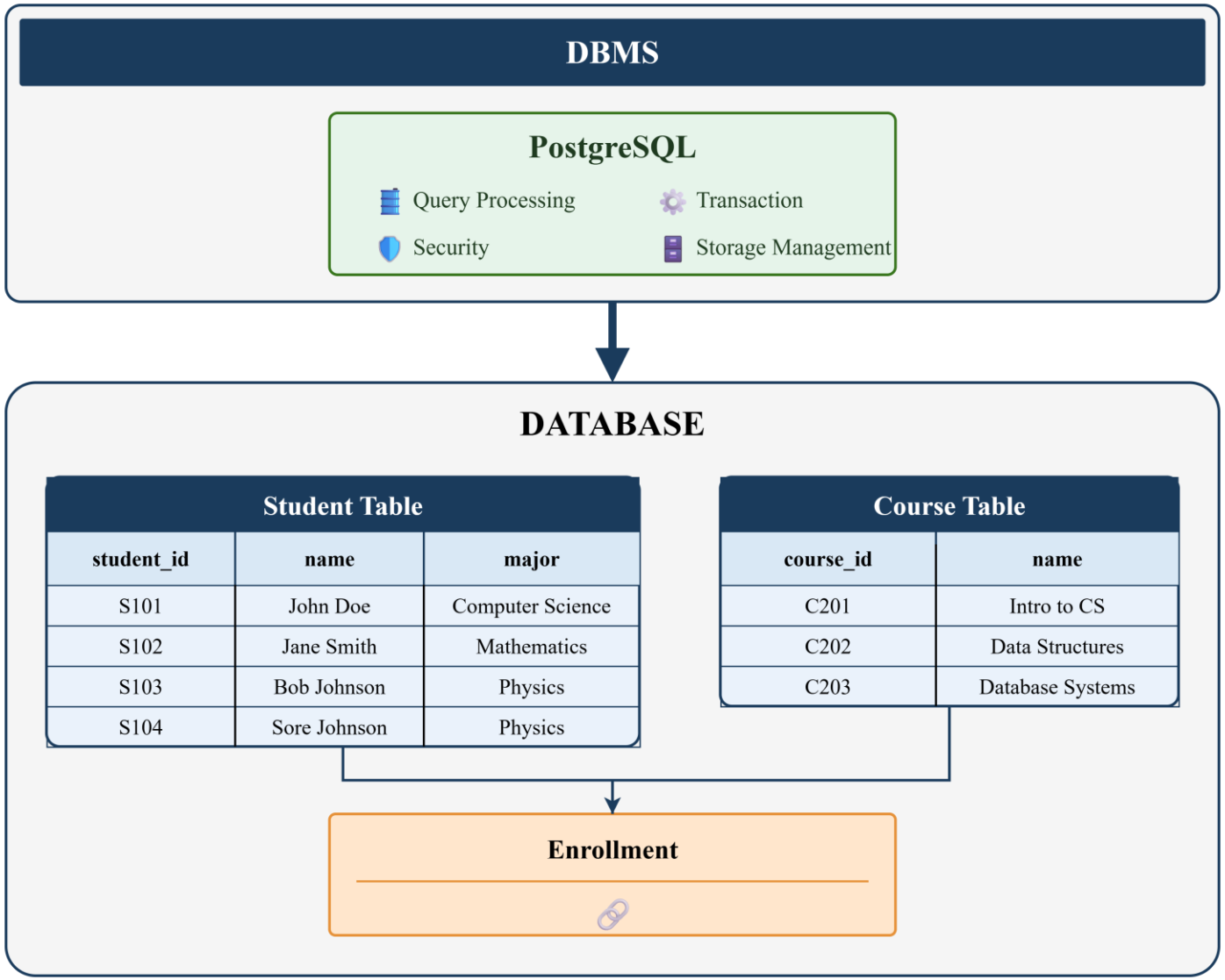
❖ Data volume, velocity and variety



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Introduction

❖ Database vs DBMS





A Database Management System (DBMS) is software that lets you create, store, manage, and retrieve data in a database

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Introduction

❖ Transaction, Index and Normalization

Transaction

A unit of work
executed as a single logical action

ACID properties

- **Atomicity** — All-or-nothing execution
- **Consistency** — Valid state to valid state
- **Isolation** — Concurrent transactions don't interfere
- **Durability** — Committed changes persist

Transaction control

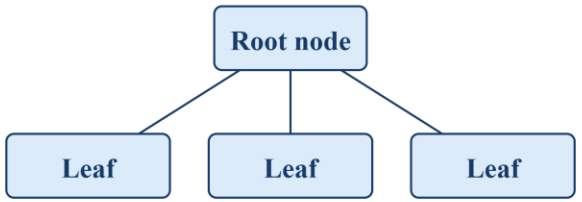
BEGIN → COMMIT / ROLLBACK

Concurrency control
Locking and isolation levels prevent conflicting access

Index

Auxiliary structure
that speeds up data retrieval

B-Tree structure



Index types

- Primary index
- Secondary index
- Unique index
- Composite index

Trade-off
Faster reads, slower writes

Normalization

Organizing data
to reduce redundancy

Progressive normal forms

1NF → 2NF → 3NF

- 1NF: Atomic column values
- 2NF: No partial dependency
- 3NF: No transitive dependency

Trade-off
More joins across related tables

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Introduction

❖ Data Warehouse













Transactions Inventory Purchasing

Collect Data

Company Departments

Marketing

Operations

Financial

Reporting







Retail Company

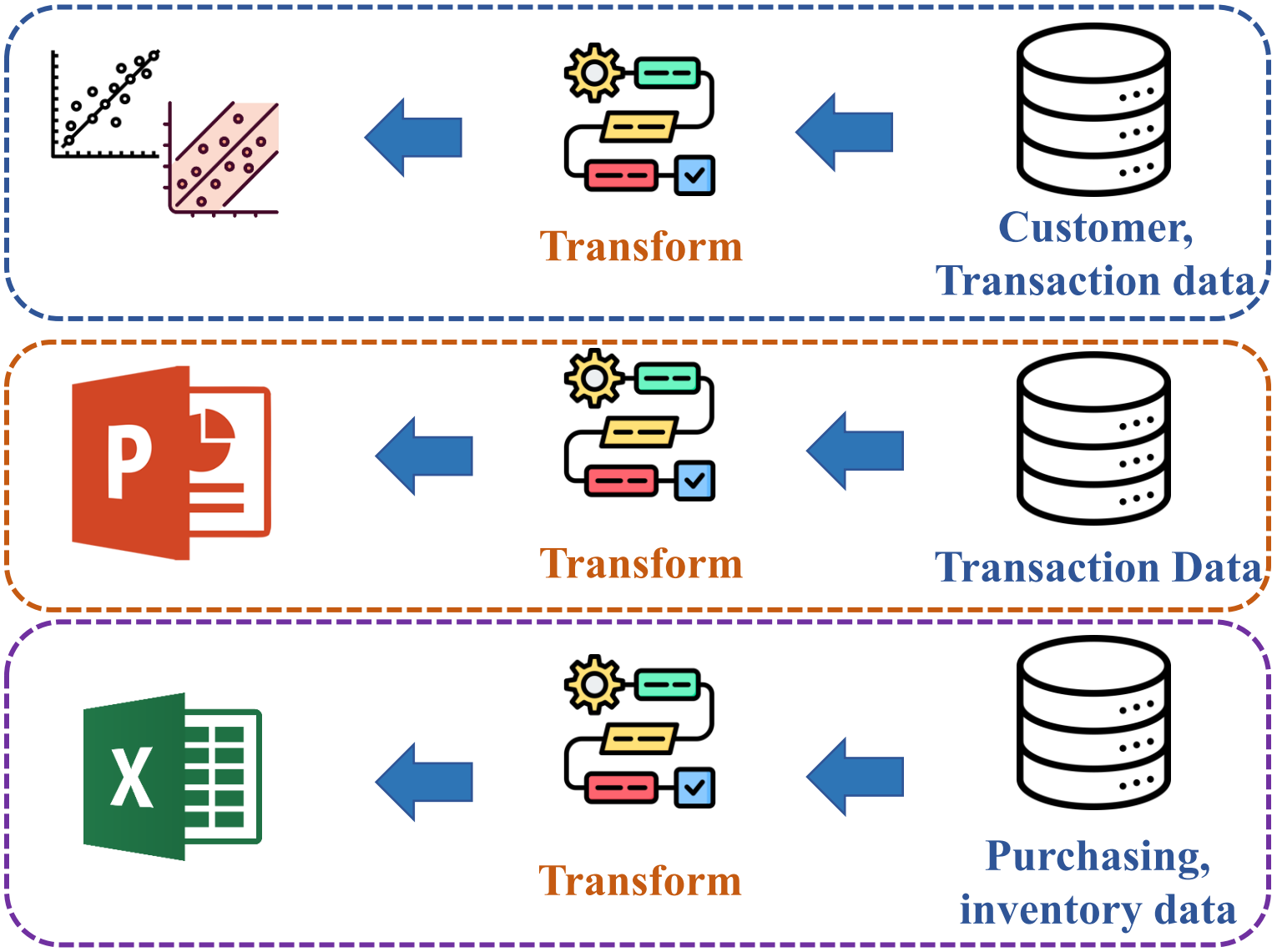
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Introduction

❖ Data Warehouse



- What was the current trends in customer behavior in the past 2 years?
- What was our total retail sales last month?
- What was the amount of inventory of key products over the last three months

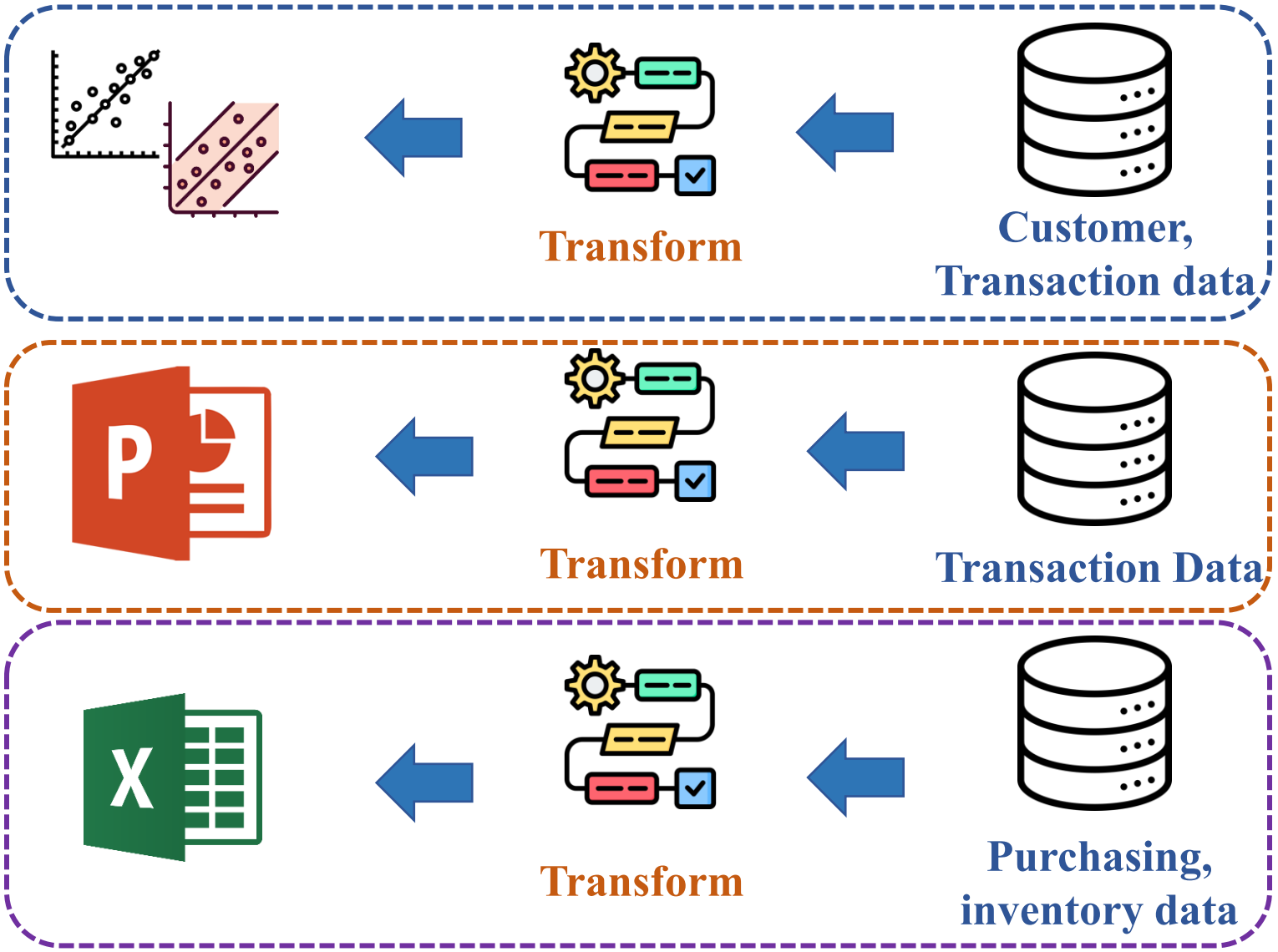


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Introduction

❖ Data Warehouse

- Time Consuming (Build report from scratch, collect data from many source)
- Many human involve (inconsistence, human error)
- Hard to have intergrated report (many tools, many terminologies,)
- Unpredicted Issues (number not match, wrong information)



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Introduction

❖ Data Warehouse



"We need a centralized, reliable, and scalable way to use our data."





Data Warehouse

Structure Data



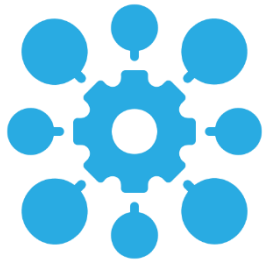
Data Lake

Semi, Unstructure Data



Data Lakehouse

Mix Warehouse and Lake



Data Mesh

Decentialize

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Introduction

❖ Data Warehouse



"We need a centralized, reliable, and scalable way to use our data."





Data Warehouse

Structure Data



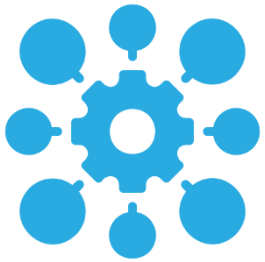
Data Lake

Semi, Unstructure Data



Data Lakehouse

Mix Warehouse and Lake



Data Mesh

Decentialize

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Introduction

❖ Data Warehouse

Multi Source Ingredients

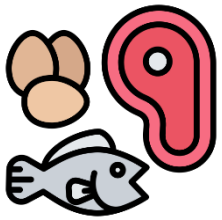
Market



Mart



Farm



Warehouse

- Collect
- Cleaning
- Processing
- Preserve
- Arrange

Kitchen

- Choose Processed Ingredients
- Cooking, Decorating
- Serving Customer

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Introduction

❖ Data Warehouse

What is a Data Warehouse?

Centralized data store
Built for reporting and analysis
Read-optimized, separate from daily operations

Purpose

Support decision-making
Consolidates data across departments
Basis for enterprise-wide reporting and analysis

Analytical vs operational

Two kinds of data

Analytical	Operational
Historical view	Current state
Read-heavy (OLAP)	Read/write (OLTP)
Cross-subject	Single process
Supports decisions	Runs daily business

Data Warehouse stores
analytical data, not operational data

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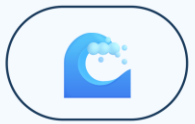
Introduction

❖ Data Lake

How is Data Lake different from Data Warehouse?

Criteria	Data Lake	Data Warehouse
Data	Raw, diverse	Processed, structured
Architecture	Storage-oriented	Subject-oriented
Purpose	Flexible, exploratory	Reporting, analysis
Users	Many data types	Structured data

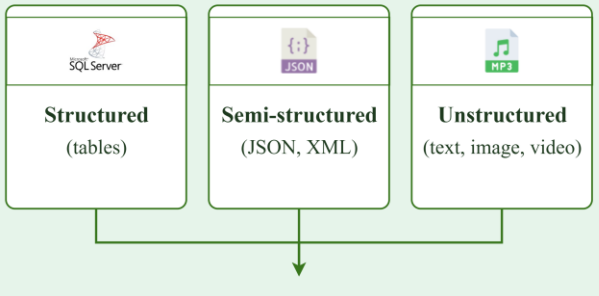
What is Data Lake?



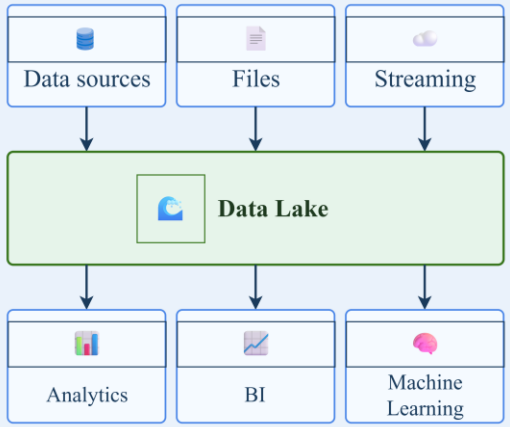
Data Lake

- Store raw data
- Large scale
- Diverse formats

Raw data



Storage-oriented architecture



When is Data Lake useful?

Diverse data
Many formats, different sources

Flexible analysis needs
Exploration, experimentation

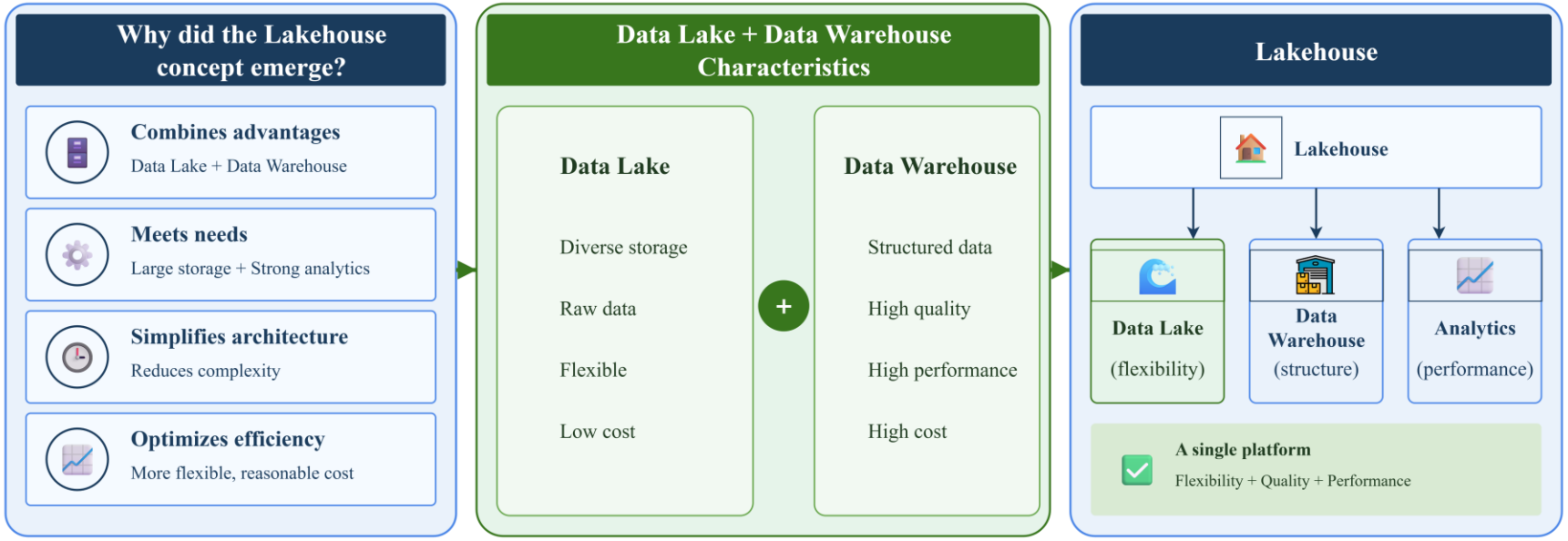
Large scale
Handle large volumes of data

Long-term storage needed
Serve multiple use cases

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Introduction

❖ Data Lakehouse



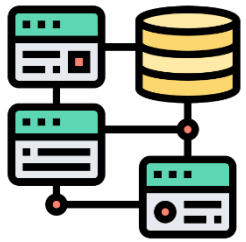
Lakehouse vs Data Warehouse vs Data Lake

Criteria	Lakehouse	Data Warehouse	Data Lake
Data	Diverse + structured	Mainly structured	Diverse (raw, semi-structured, unstructured)
Architecture	Unified (storage + analytics)	Separate (dedicated data warehouse)	Raw storage
Use cases	Need both flexibility and performance	Traditional reporting, analysis	Store large data, explore
Trade-off	Balance cost, performance, flexibility	High performance, less flexible	High flexibility, data quality needs more processing

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Introduction

❖ Data Warehouse & ETL





mongoDB.

Multi Sources

Extract



Same Format Data

Transform





Processed Data

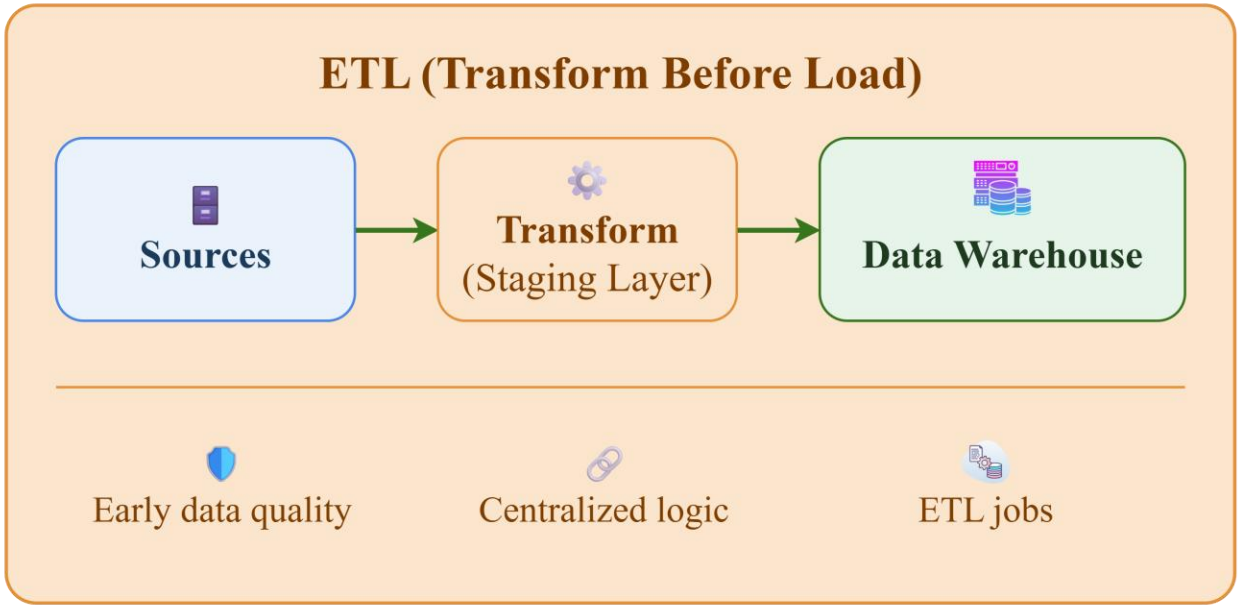
Loading

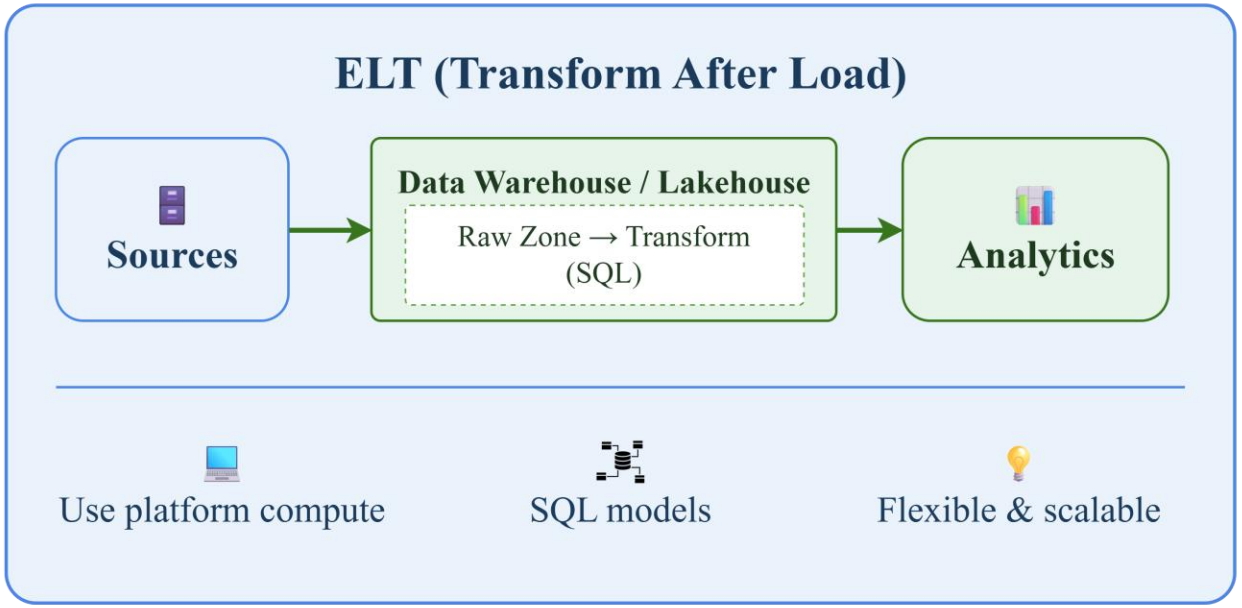


Data Warehouse

Introduction

❖ ETL vs ELT





ETL and ELT are both data integration methods, but the core difference lies in when the **transformation step** occurs: ETL transforms data **before** loading it into the target system, whereas ELT loads all raw data into the target system and then performs the transformation directly there, leveraging modern storage and computing infrastructure.

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Database to Data Warehouse

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Database to Data Warehouse

❖ What is the purpose of an operational database?

Station 8767
Urban Monitoring Station

Time	AQI	PM2.5	Temp (°C)
19:00	102	102	27.4
20:00	108	108	27.1
21:00	111	111	26.8

Station 8767
Suburban Monitoring Station

Time	AQI	PM2.5	Temp (°C)
19:00	56	56	24.1
20:00	58	58	23.9
21:00	60	60	23.7

Station 8767
Industrial Area Station

Time	AQI	PM2.5	Temp (°C)
19:00	162	162	30.1
20:00	158	158	29.6
21:00	155	155	29.2

INSERT measurement

INSERT measurement

INSERT measurement

Operational Database in Daily Operation

Table: Measurements

Field	Type / Key
station_id	(FK)
measured_at	(datetime)
aqi	(number)
pm25	(number)
temperature	(number)

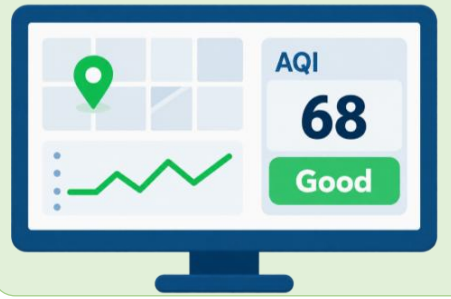
Table: Station info

Field	Type / Key
station_id	(PK)
station_name	(text)
location	(text)
status	(text)

SELECT current data

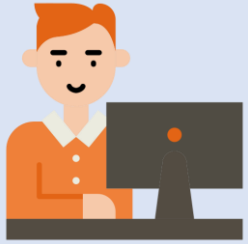
UPDATE station info

Application / API
Provides current air-quality data to users and other systems.



End users
Web / Mobile / Other systems

Admin / Operator
Manages station information and system settings.



- Update station status
- Edit location
- Maintain device info
- Add or remove stations

An operational database supports the day-to-day operation of the application by processing many small inserts, reads, and updates.

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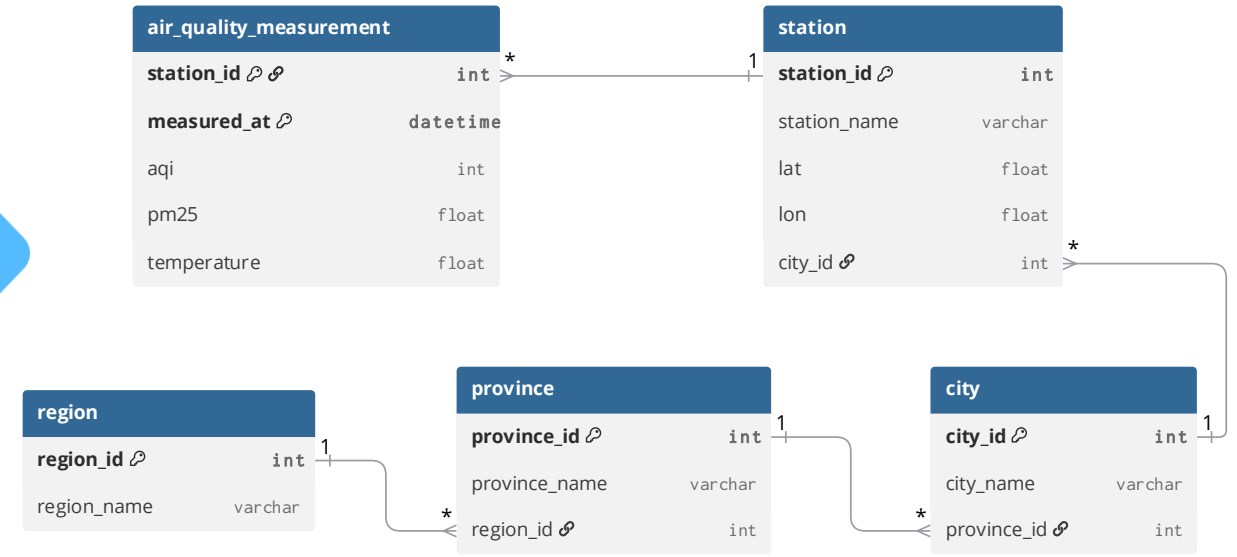
Database to Data Warehouse

❖ When the need shifts from operations to analytics

station_id	station_name	lat	lon	city	measured_at	aqi	pm25	temperature
8767	HCMC US Consulate	10.782978	106.700711	Ho Chi Minh City	2021-01-21 19:00	102	102	27.4
8767	HCMC US Consulate	10.782978	106.700711	Ho Chi Minh City	2021-01-21 20:00	108	108	27.1
8641	Hanoi US Embassy	21.021506	105.818875	Hanoi	2021-01-21 20:00	281	281	18.5

Station information is repeated in every measurement → If a station's information changes (e.g. location), all related rows need to be updated.

Normalization



- Less data duplication
- Easier and consistent updates
- Clearer data structure
- Suitable for transactional (operational) workload

Operational Database usually organizes data in a normalized model (e.g. 3NF) so that it is convenient to insert, update, and maintain data consistently.

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Database to Data Warehouse

❖ When the need shifts from operations to analytics

1. Stations Generate Data

Stations continuously send measurements.

Station 8767
Urban Monitoring Station

Time	AQI	PM2.5	Temp (°C)
19:00	102	102	27.4
20:00	108	108	27.1

Station 8767
Suburban Monitoring Station

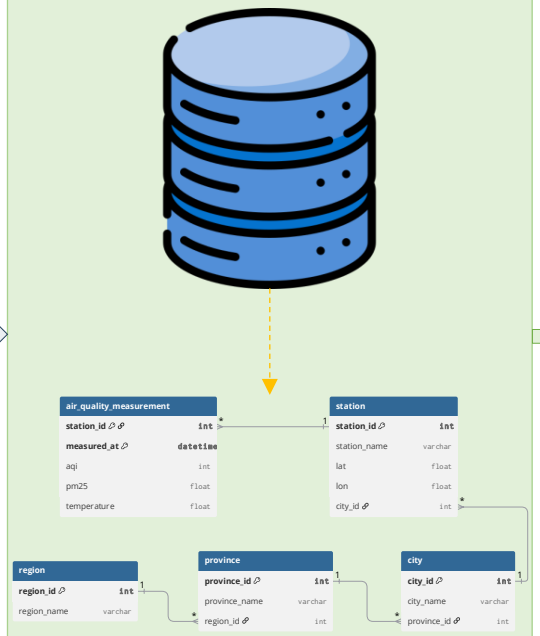
Time	AQI	PM2.5	Temp (°C)
19:00	56	56	24.1
20:00	58	58	23.9

Station 8767
Industrial Area Station

Time	AQI	PM2.5	Temp (°C)
19:00	162	162	30.1
20:00	158	158	29.6

2. Operational Database Stores Data

Measurements are continuously inserted.



3. Scheduled Analytics Processing

At a scheduled time, data is read and processed.

- Nightly schedule
- **Filter** — Select relevant data
- **Group** — Organize by time, location...
- **Aggregate** — Calculate summaries

4. Morning Reporting

Dashboard / Report

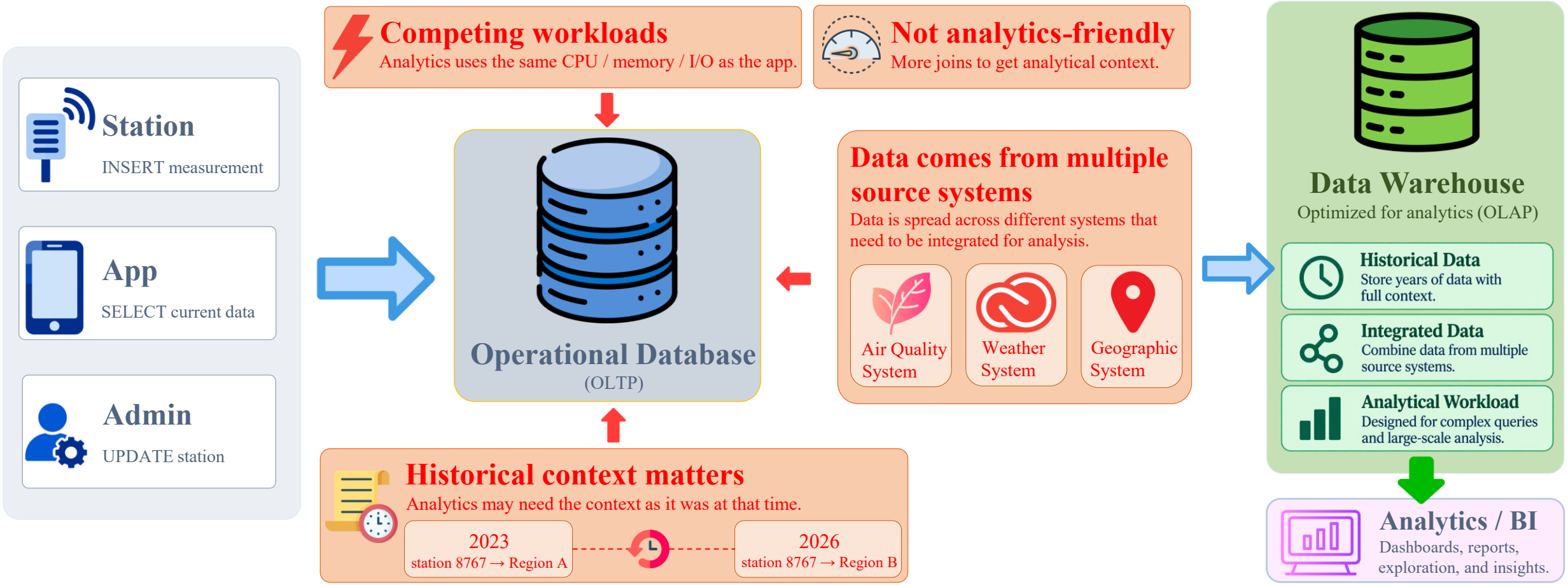


As data is stored over time, the questions become analytical and can be answered through scheduled processing and reporting.

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Database to Data Warehouse

❖ Why not analyze directly on the Operational Database?

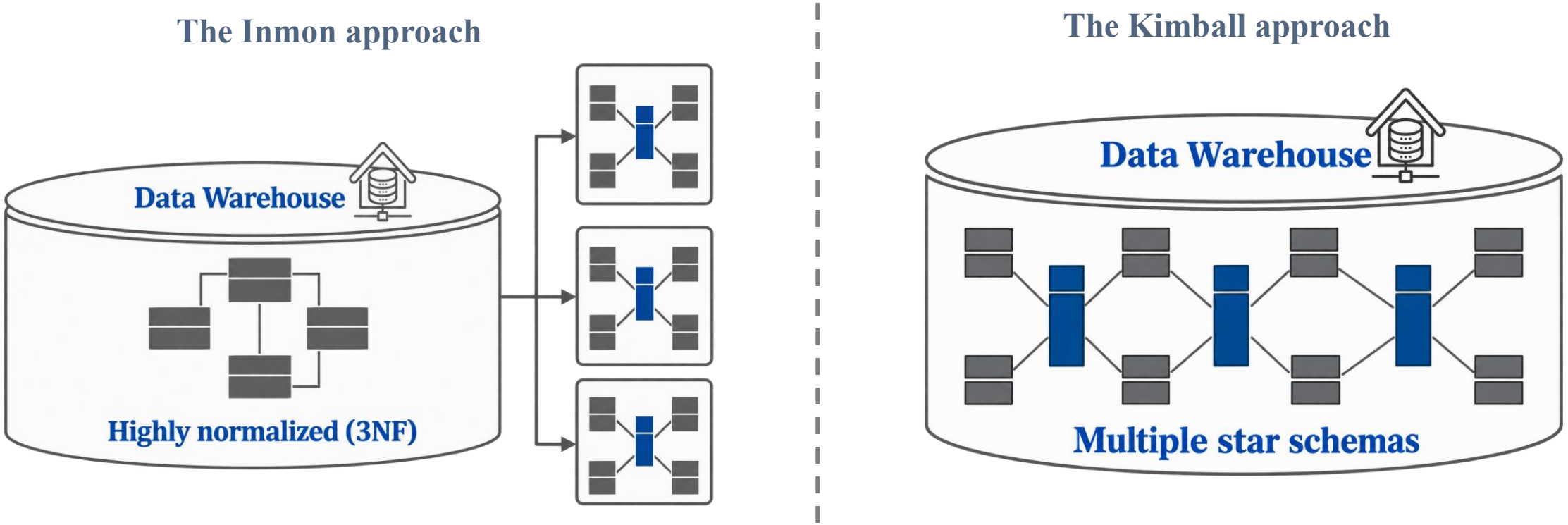


Operational Database runs the application. Data Warehouse supports large-scale, integrated, historical analytics.

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Database to Data Warehouse

❖ How should data in a data warehouse be organized?

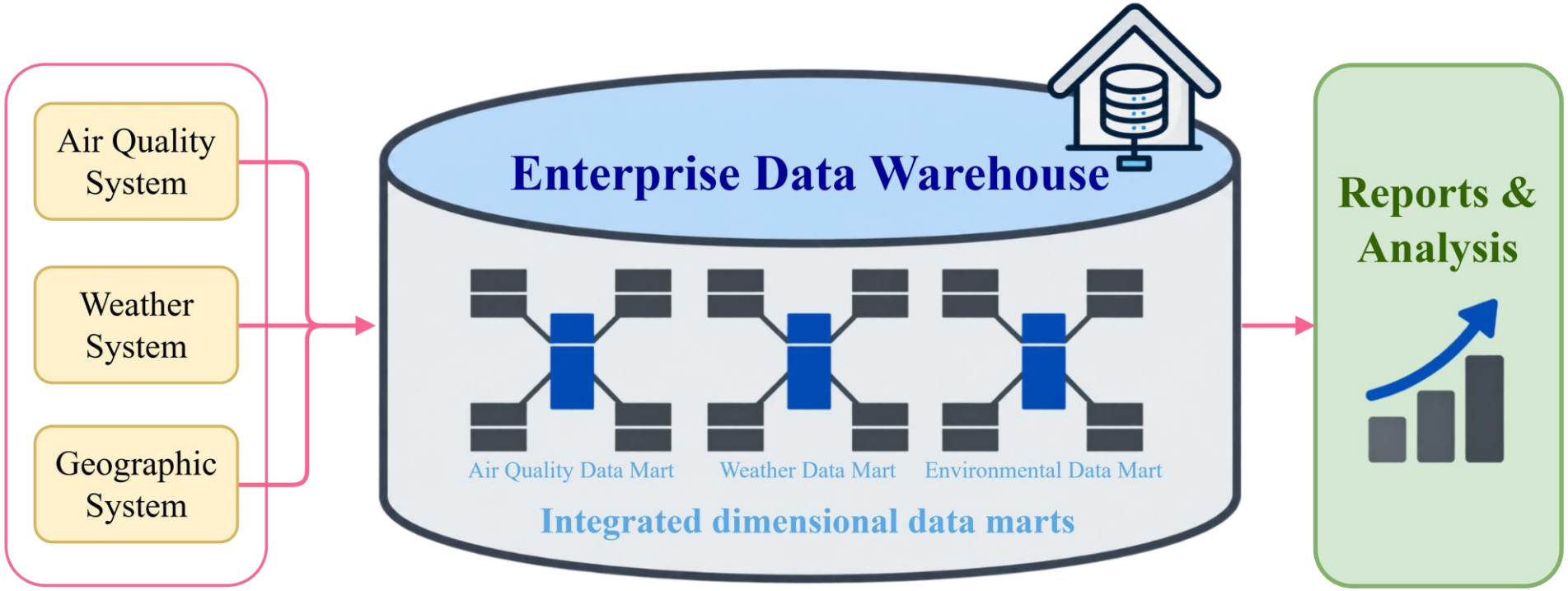


Inmon and Kimball offer two different ways to organize a data warehouse: normalized 3NF vs dimensional star schemas.

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Database to Data Warehouse

❖ Inmon Data Modeling Approach (Top-down approach)

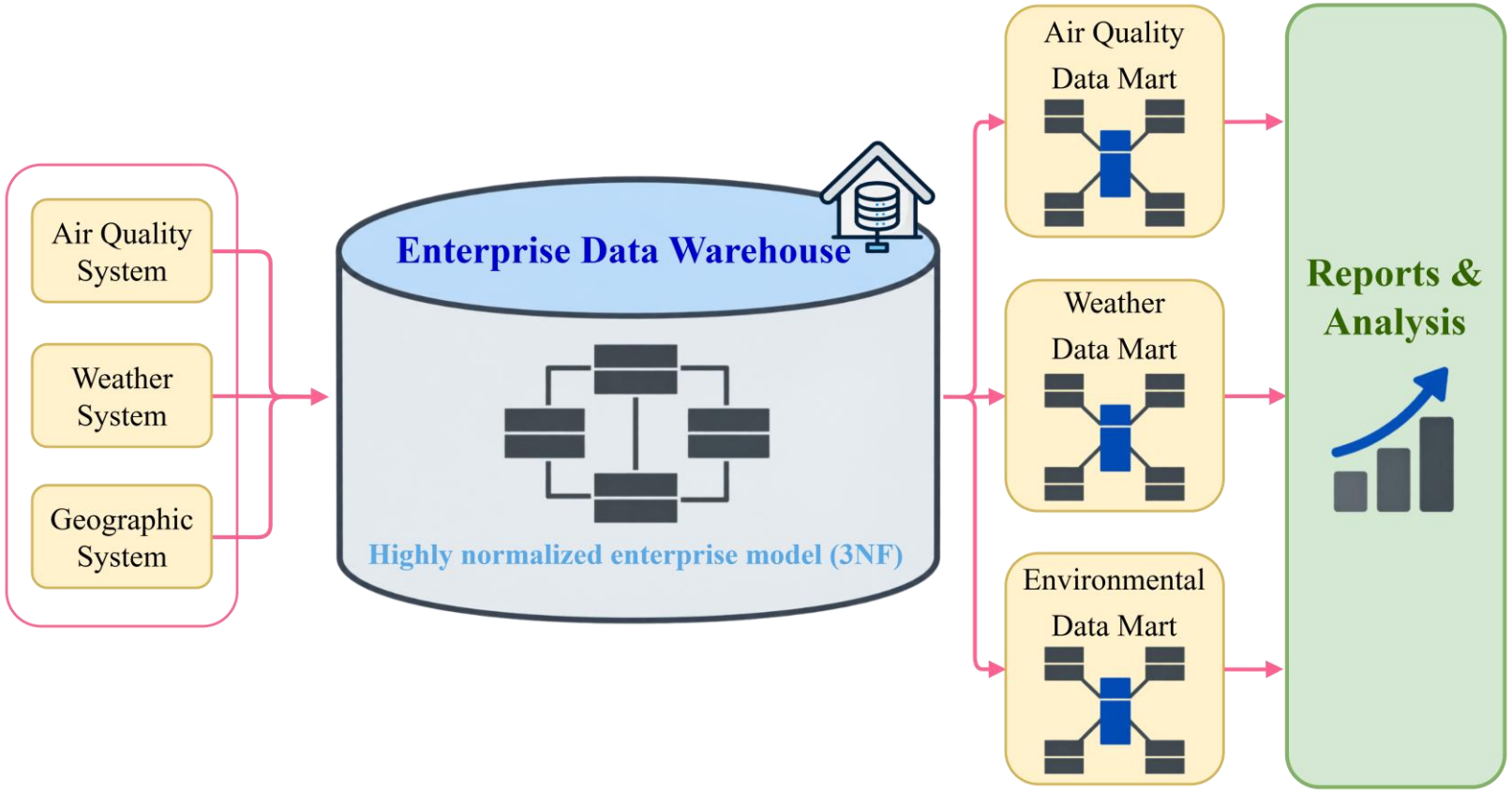


Build an integrated, normalized enterprise data warehouse first, then derive data marts for specific analytical needs.

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Database to Data Warehouse

❖ Kimball Data Modeling Approach (Bottom-up approach)



Build dimensional data marts around business processes first, then integrate them into an enterprise-wide warehouse.

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Database to Data Warehouse

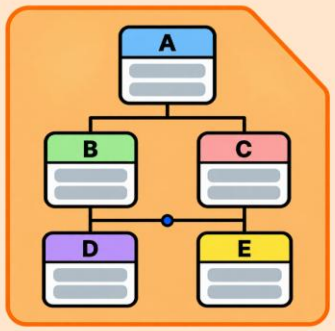
❖ Inmon vs. Kimball: Key Differences

Aspect	Inmon Approach	Kimball Approach
Starting Point	Build a common data foundation for the entire enterprise	Start from individual business processes with analytical needs
Development Direction	Top-down	Bottom-up
Initial Data Foundation	Enterprise Data Warehouse, typically normalized	Data Marts organized by business process
Data Marts	Derived from the EDW	Built incrementally but designed to integrate with each other
Scaling Approach	Expand from enterprise-wide data to specific analytics areas	Expand from individual business processes toward an enterprise-wide warehouse
Advantage	Enterprise-wide consistency is designed from the beginning	Delivers value incrementally without waiting for a complete enterprise model
Limitation	Requires more upfront modeling and effort before the first Data Marts are delivered	Requires strong governance and integration standards to keep Data Marts consistent as they grow

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Database to Data Warehouse

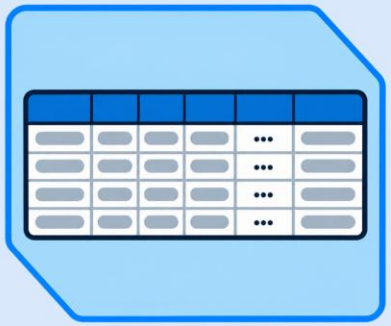
❖ From Data Warehouse Architecture to Analytical Data Model



Normalized Model

Multiple related tables

- Data is split into multiple tables to reduce redundancy
- Tables are connected through relationships
- Common in operational systems and sometimes used in EDW

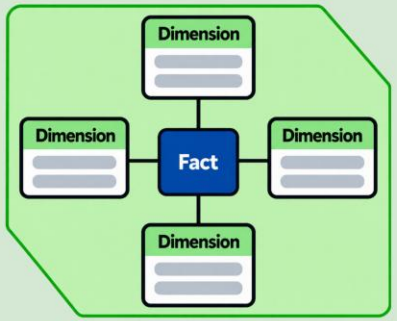


One Big Table

One wide analytical table

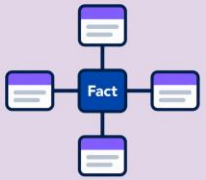
col_1	col_2	col_3	...	col_n
			...	
			...	
			...	

- All relevant attributes are stored in a single wide table
- Simple to query
- Can be inefficient due to data redundancy and many sparse columns



Dimensional Model

Facts and dimensions



Star Schema

Dimensions connect directly to the central fact table



Snowflake Schema

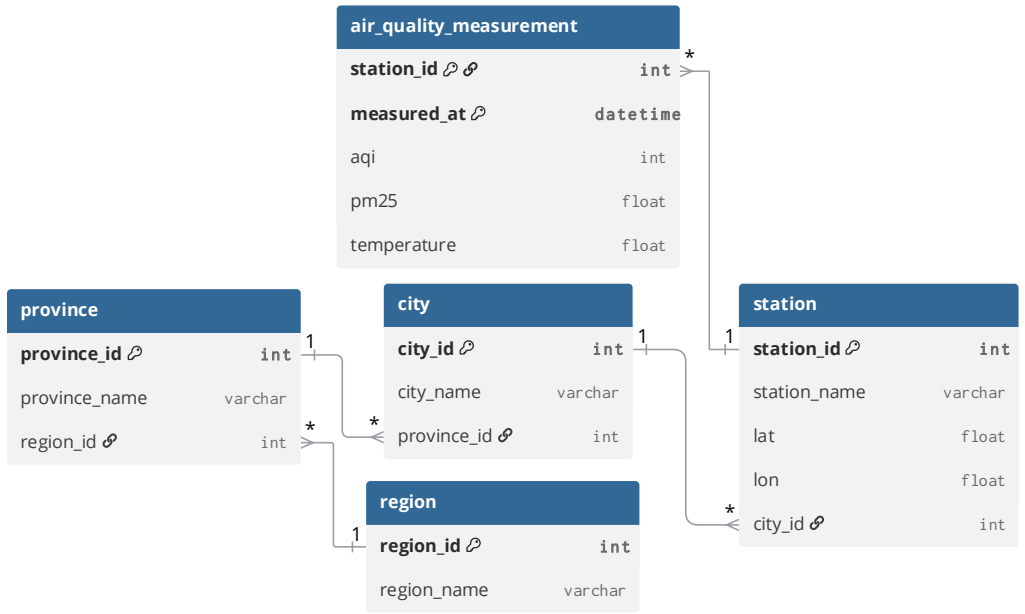
Dimensions are further normalized into related tables

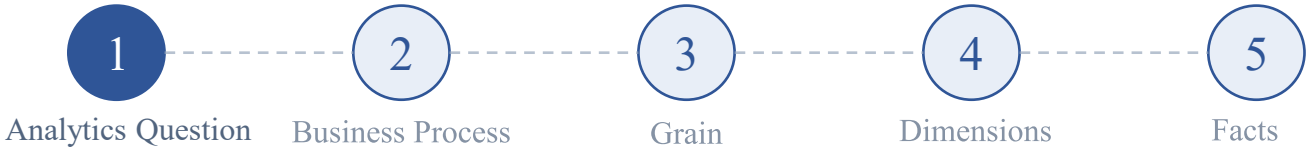
Analytical data can be organized in different ways, from normalized relational models to wide tables and dimensional schemas, depending on the intended workload and trade-offs.

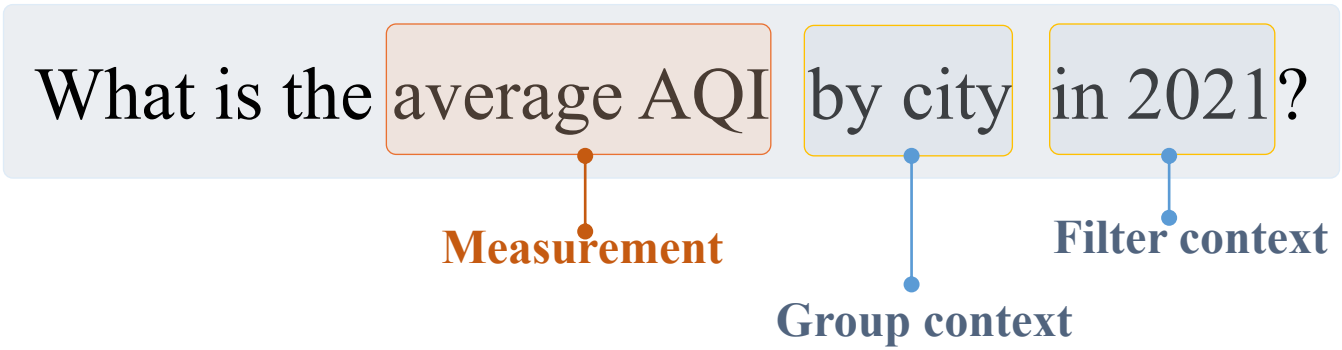
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Database to Data Warehouse

❖ View analytics questions through the lenses of measurement and context.





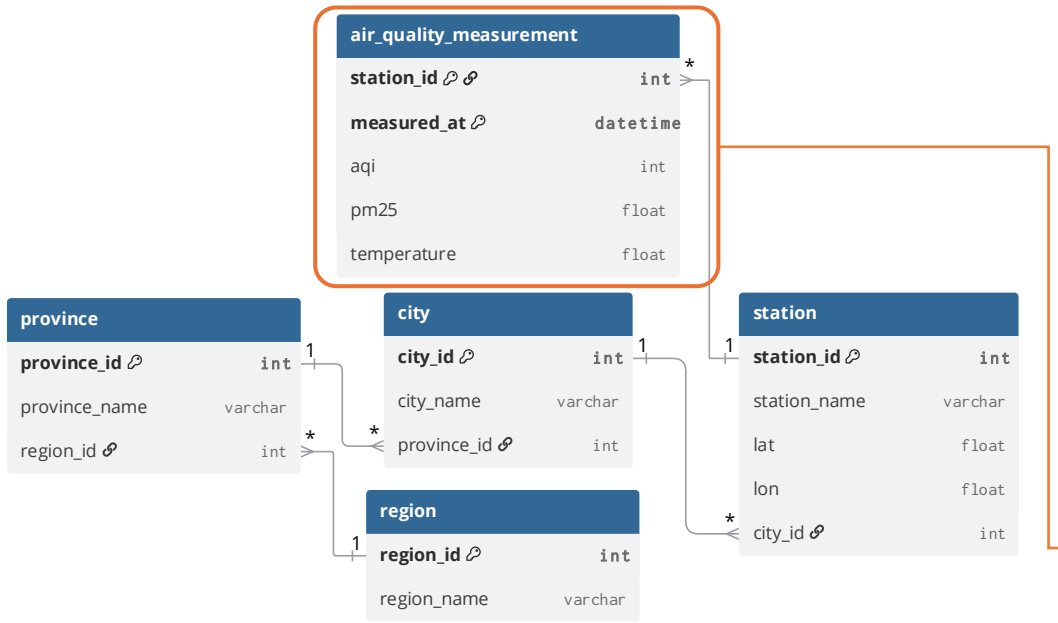


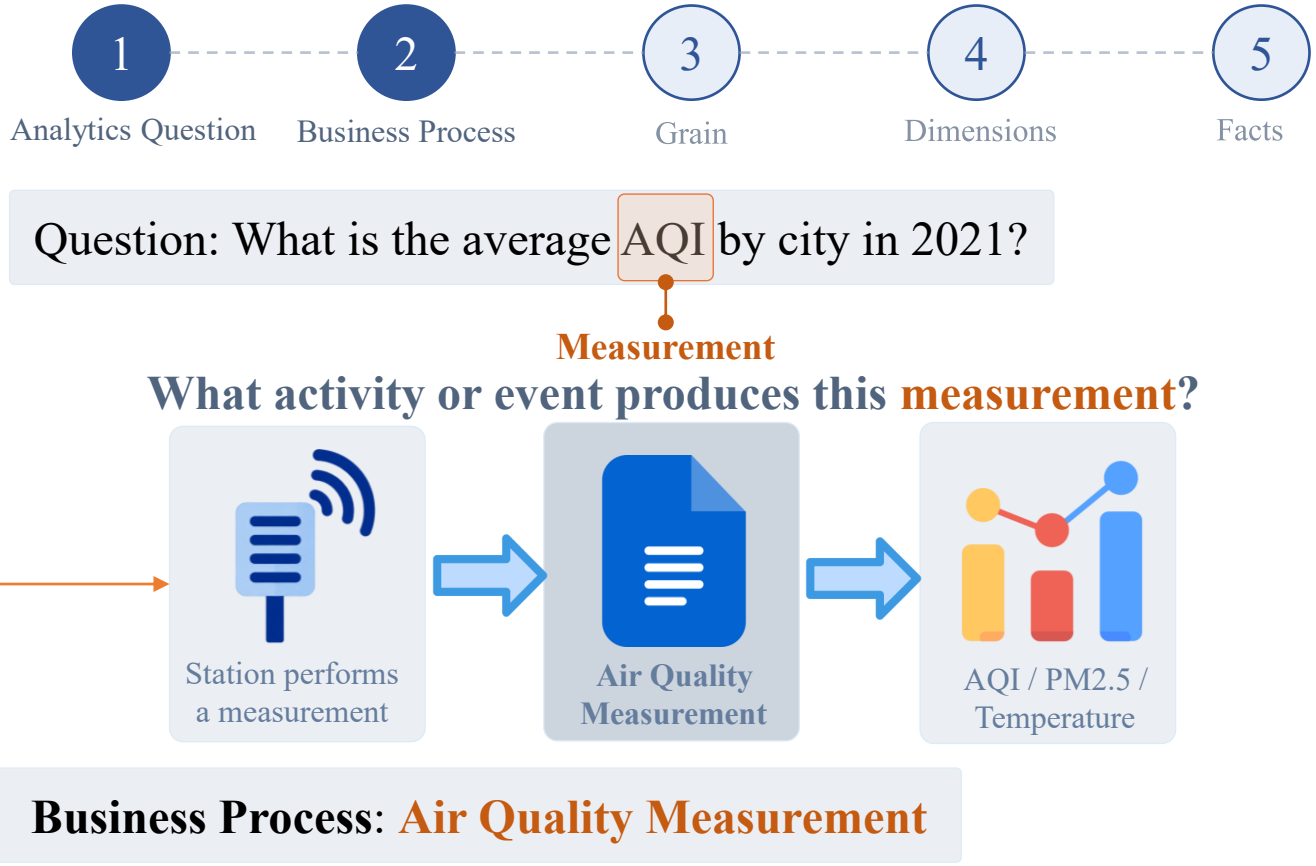
Before designing a star schema, we should first identify what needs to be measured and in what context the user wants to analyze it.

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Database to Data Warehouse

❖ Identifying the Business Process



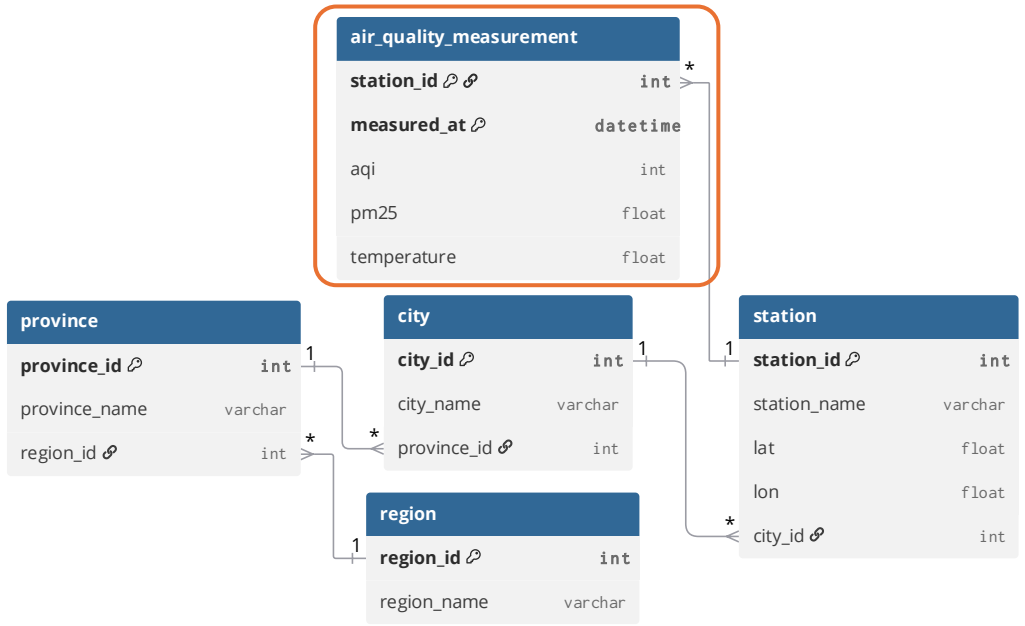


A Business Process is the activity or event that generates the measurements we want to analyze.

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Database to Data Warehouse

❖ Grain



1 Analytics Question — 2 Business Process — 3 Grain — 4 Dimensions — 5 Facts

Business Process: Air Quality Measurement

How should one row be represented?

Daily summary

1 row = one station + one day

station_id	date	avg_aqi
8767	2021-01-21	107

- keeps daily-level analysis
- loses within-day detail

Atomic grain

1 row = one station + one measurement time

station_id	measured_at	aqi
8767	2021-01-21 19:00	102
8767	2021-01-21 20:00	108
8767	2021-01-21 21:00	111

- keeps the lowest useful detail
- can still be aggregated later

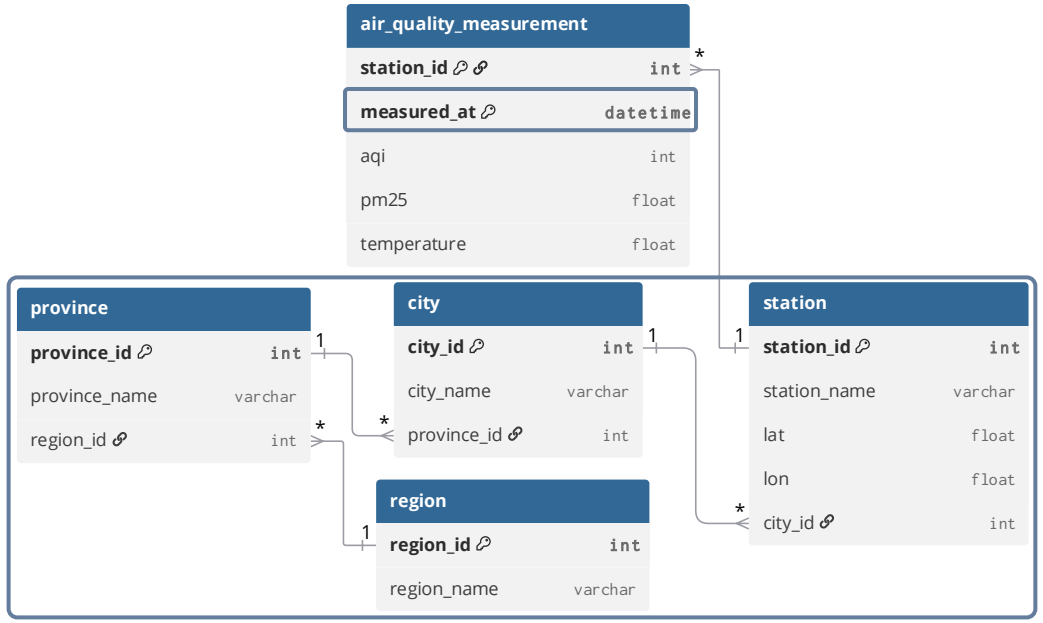
Chosen Grain = one Air Quality Measurement at one station at one point in time

Choosing the grain means deciding what one row represents, and in dimensional modeling it is often best to keep the lowest useful level of detail.

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Database to Data Warehouse

❖ Identifying Dimensions



1 Analytics Question — 2 Business Process — 3 Grain — 4 Dimensions — 5

Business Process: Air Quality Measurement

Chosen Grain = one Air Quality Measurement at one station at one point in time

What context do we need to describe each measurement?

DIM_STATION	
station_key (PK)	→ surrogate key
station_id	→ source business key
station_name	
lat	
lon	
city	
province	
region	

DIM_DATE
date_key (PK)
date
day
month
quarter
year

DIM_TIME
time_key (PK)
hour
minute
time_of_day

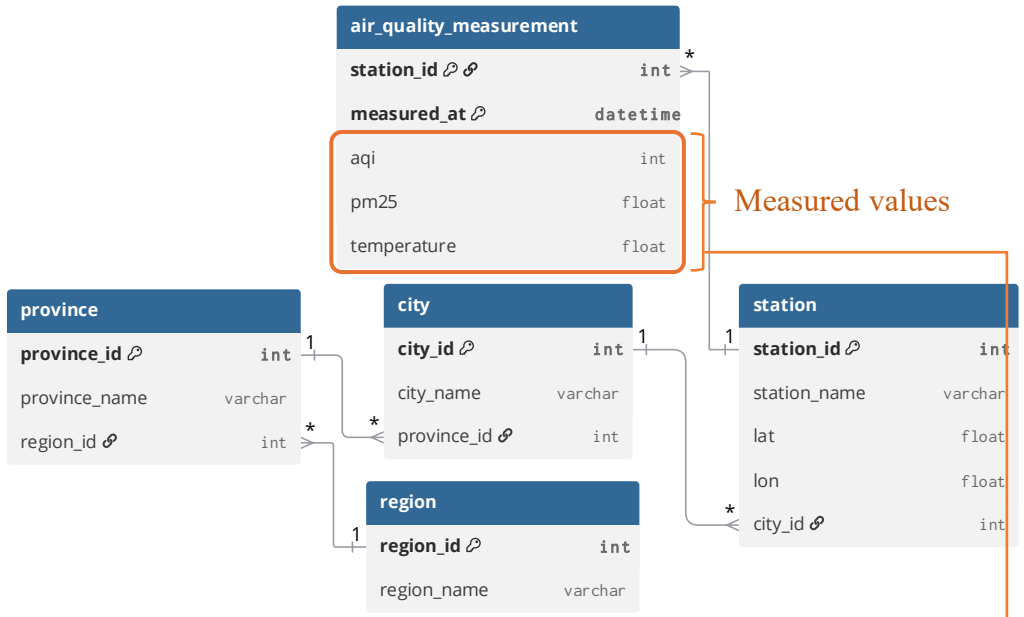
derived from measured_at

To identify dimensions, we look for the descriptive context needed for analysis and reorganize that context from the source data into analytical dimension tables.

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Database to Data Warehouse

❖ Identifying Facts



1 Analytics Question — 2 Business Process — 3 Grain — 4 Dimensions — 5

Business Process: Air Quality Measurement

Chosen Grain = one Air Quality Measurement at one station at one point in time

What values are measured at the chosen grain?

FACT_AIR_QUALITY_MEASUREMENT	
Linking keys	station_key (FK)
	date_key (FK)
	time_key (FK)
Facts	aqi
	pm25
	temperature

Facts are the quantitative measurements recorded at the chosen grain, while the Fact Table also carries the keys that connect those measurements to their analytical context.

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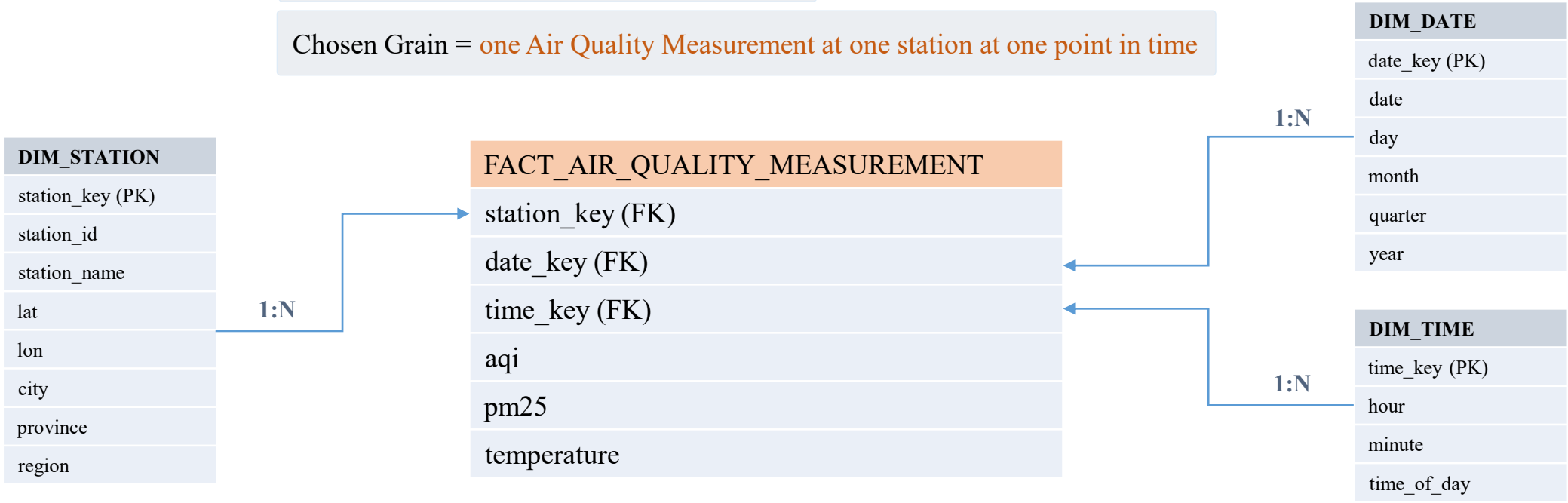
Database to Data Warehouse

❖ Final Star Schema

1 Analytics Question — 2 Business Process — 3 Grain — 4 Dimensions — 5 Facts

Business Process: Air Quality Measurement

Chosen Grain = one Air Quality Measurement at one station at one point in time

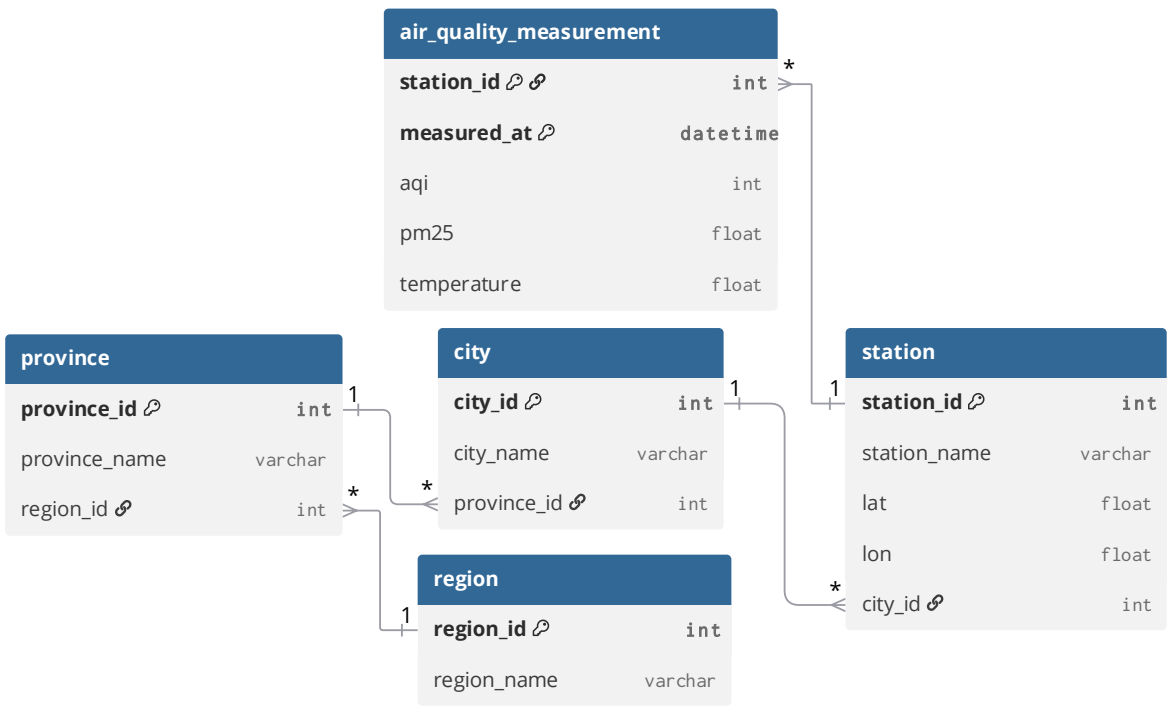


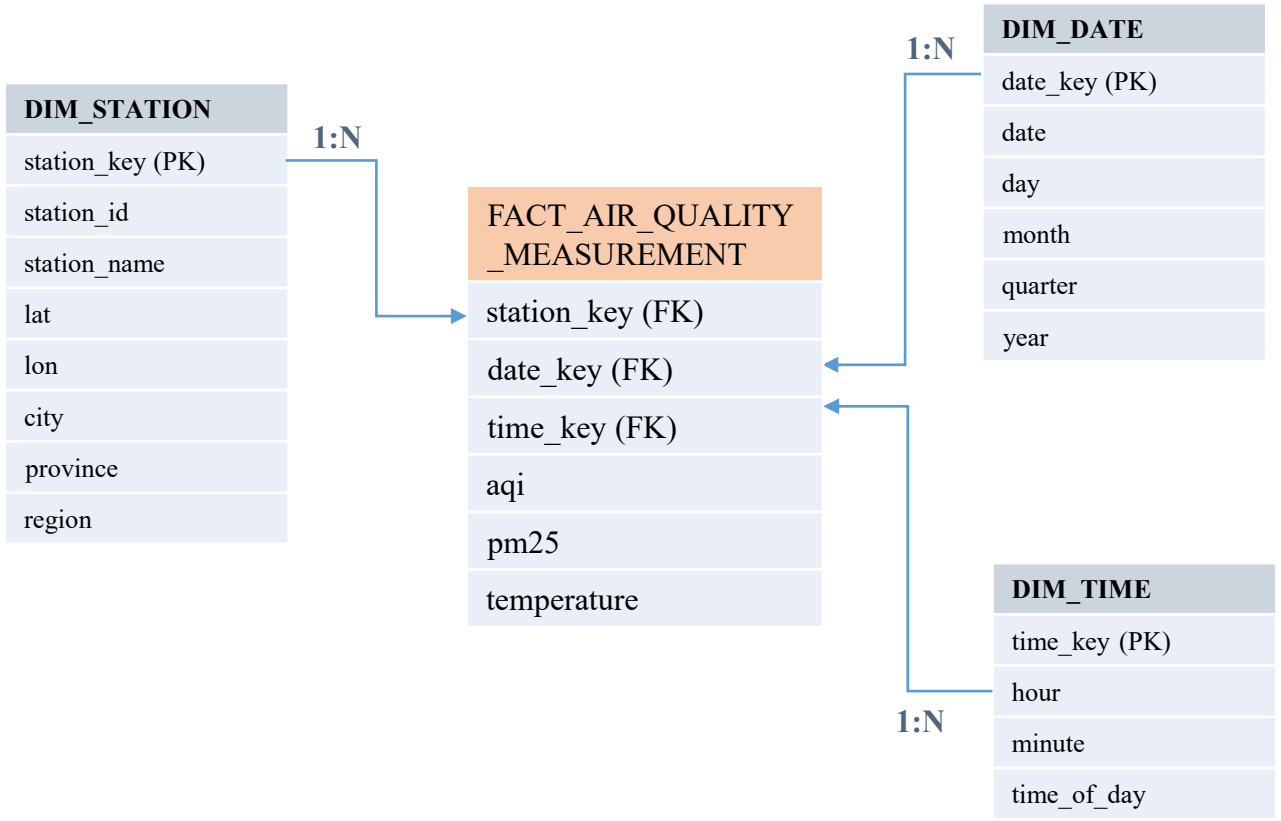
The final Star Schema is built from the business process, declared grain, analytical dimensions, and measurable facts.

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Database to Data Warehouse

❖ How Star Schema Changes Querying





Choosing the grain means deciding what one row represents, and in dimensional modeling it is often best to keep the lowest useful level of detail.

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Database to Data Warehouse

❖ How Star Schema Changes Querying

What was the average AQI by region and month in 2021?

Normalized Model

```
SELECT
    r.region_name,
    EXTRACT(MONTH FROM m.measured_at) AS month,
    AVG(m.aqi) AS avg_aqi
FROM AIR_QUALITY_MEASUREMENT m
JOIN STATION s
    ON m.station_id = s.station_id
JOIN CITY c
    ON s.city_id = c.city_id
JOIN PROVINCE p
    ON c.province_id = p.province_id
JOIN REGION r
    ON p.region_id = r.region_id
WHERE EXTRACT(YEAR FROM m.measured_at) = 2021
GROUP BY
    r.region_name,
    EXTRACT(MONTH FROM m.measured_at);
```

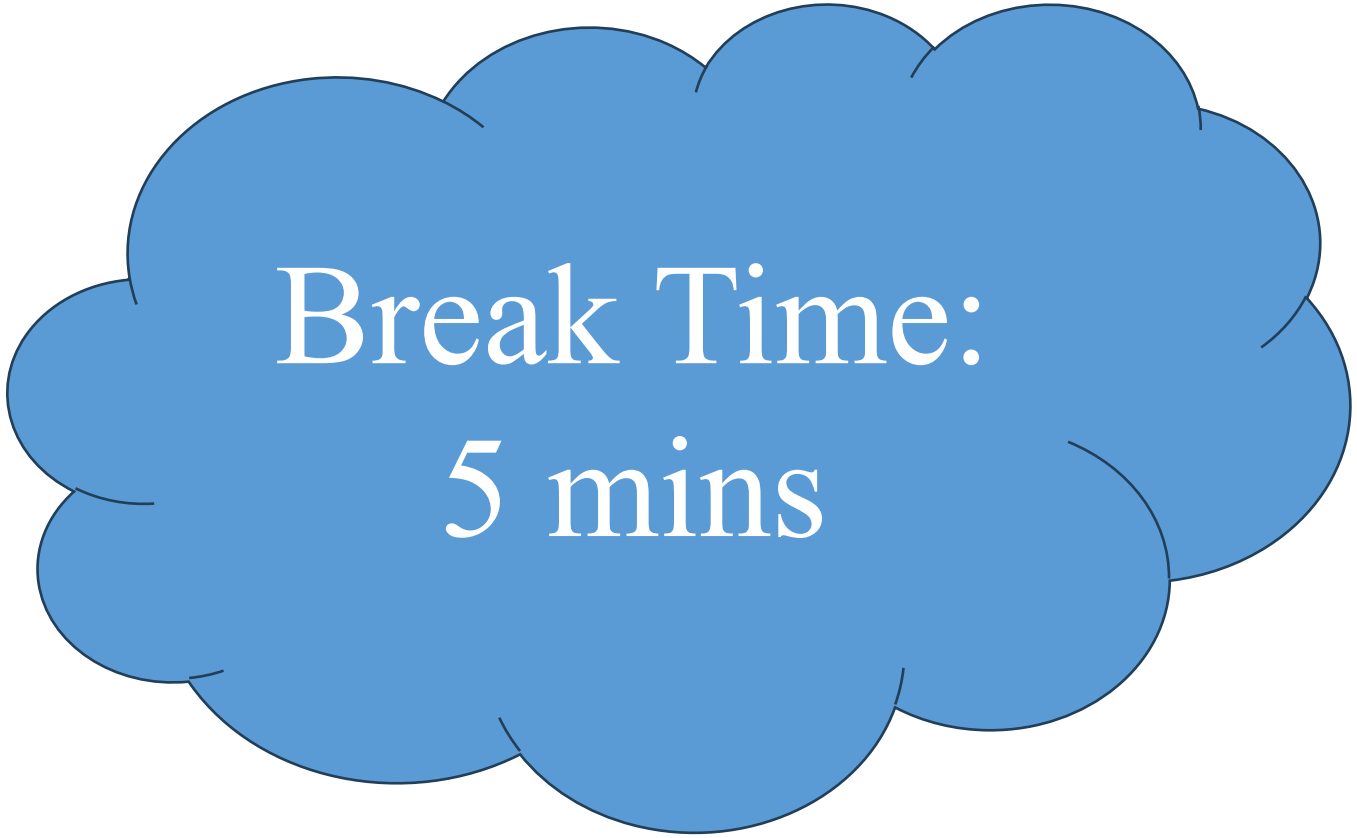
Star schema

```
SELECT
    s.region,
    d.month,
    AVG(f.aqi) AS avg_aqi
FROM FACT_AIR_QUALITY_MEASUREMENT f
JOIN DIM_STATION s
    ON f.station_key = s.station_key
JOIN DIM_DATE d
    ON f.date_key = d.date_key
WHERE d.year = 2021
GROUP BY
    s.region,
    d.month;
```

Star Schema makes analytical queries more direct by placing measurements in Facts and commonly used analytical context in Dimensions.

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Break

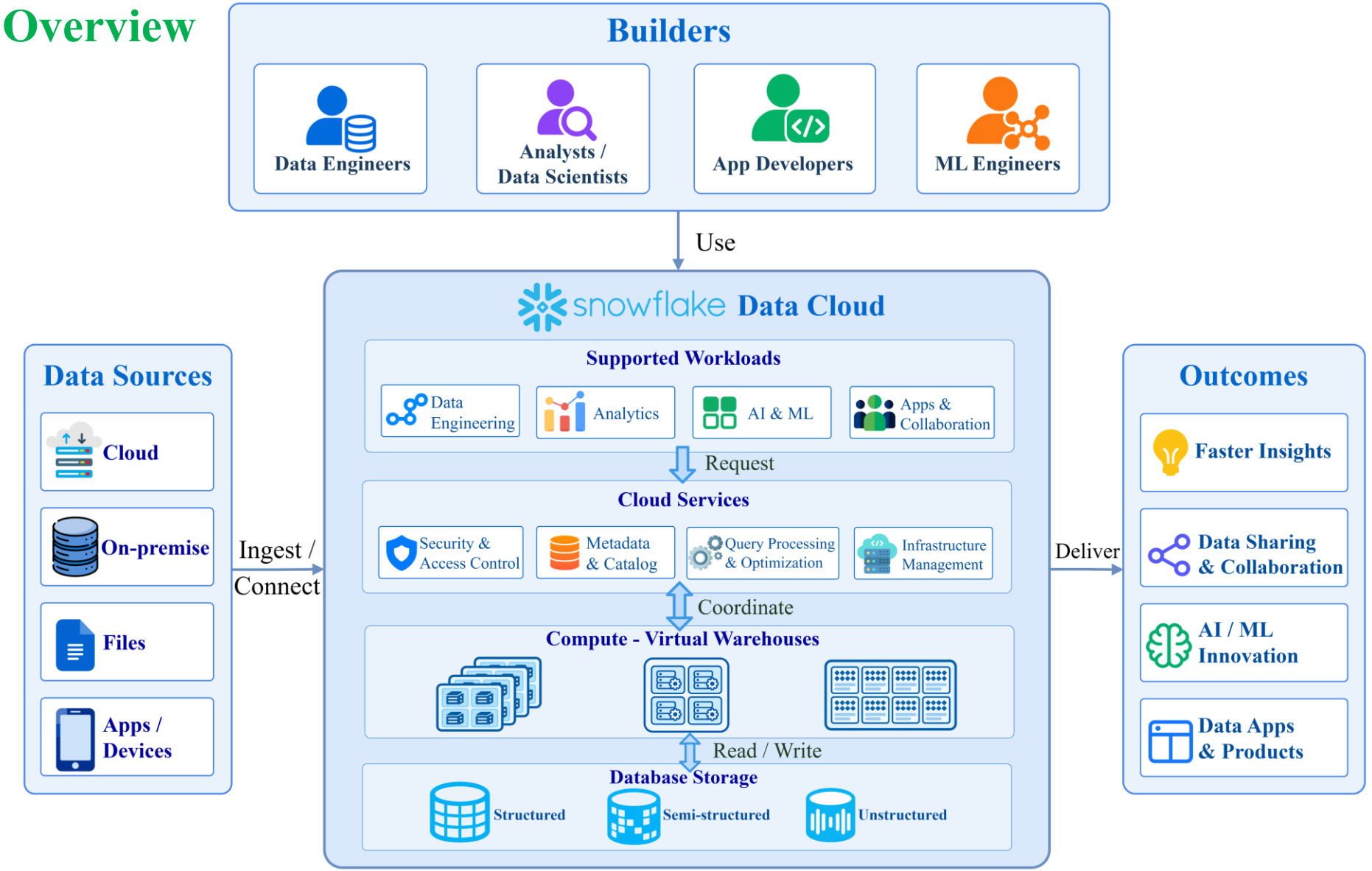


Snowflake

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Snowflake

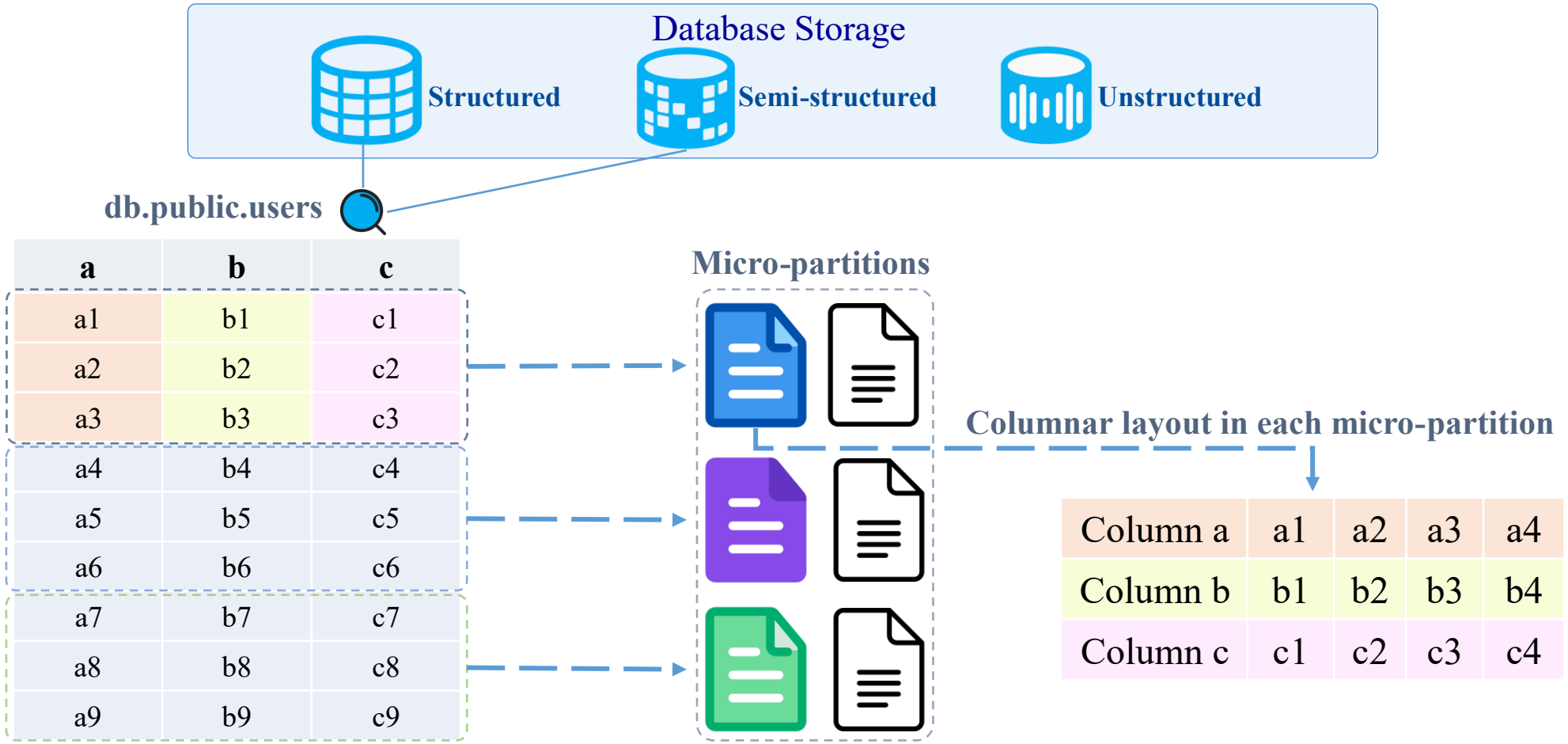
❖ Snowflake Overview



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Snowflake

❖ Snowflake Architecture: Database Storage

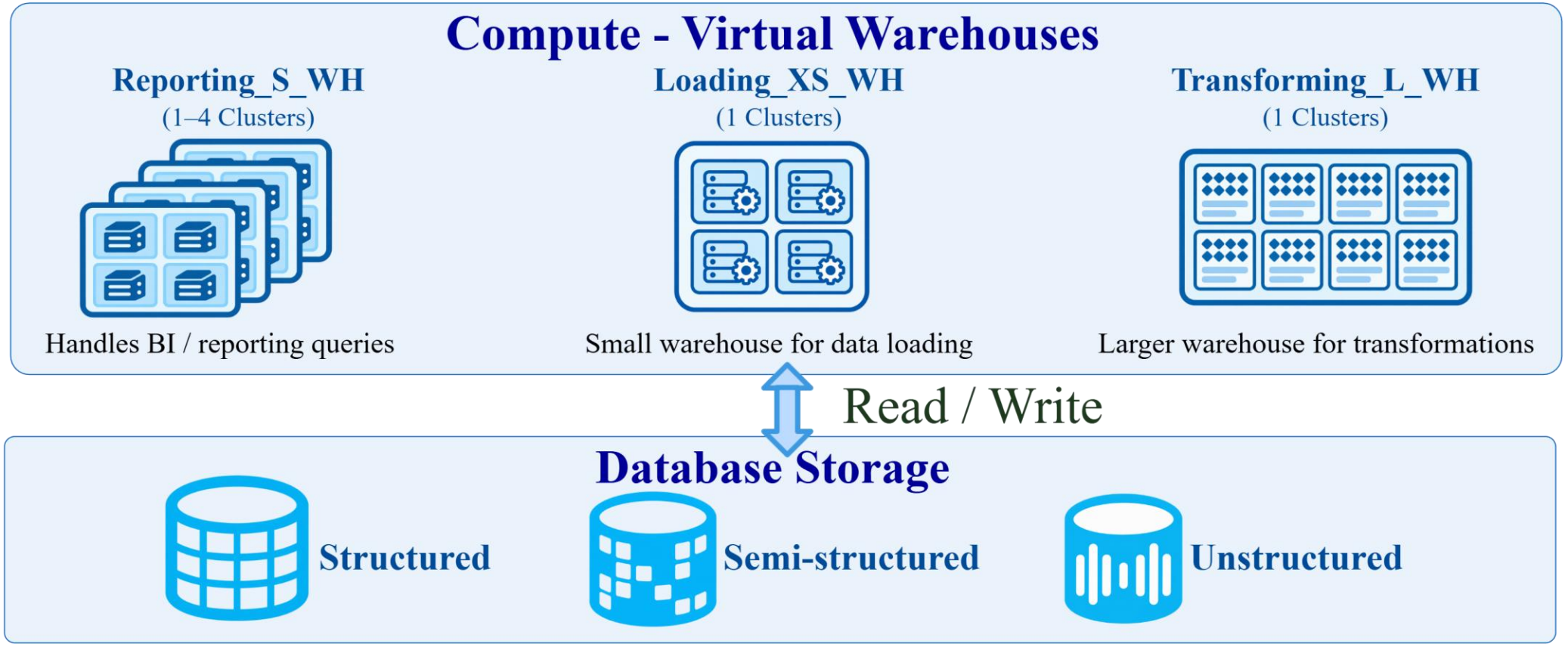


Database Storage keeps data persistent in Snowflake. Table data is automatically organized into columnar micro-partitions, while compute remains separate for processing.

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Snowflake

❖ Snowflake Architecture: Compute

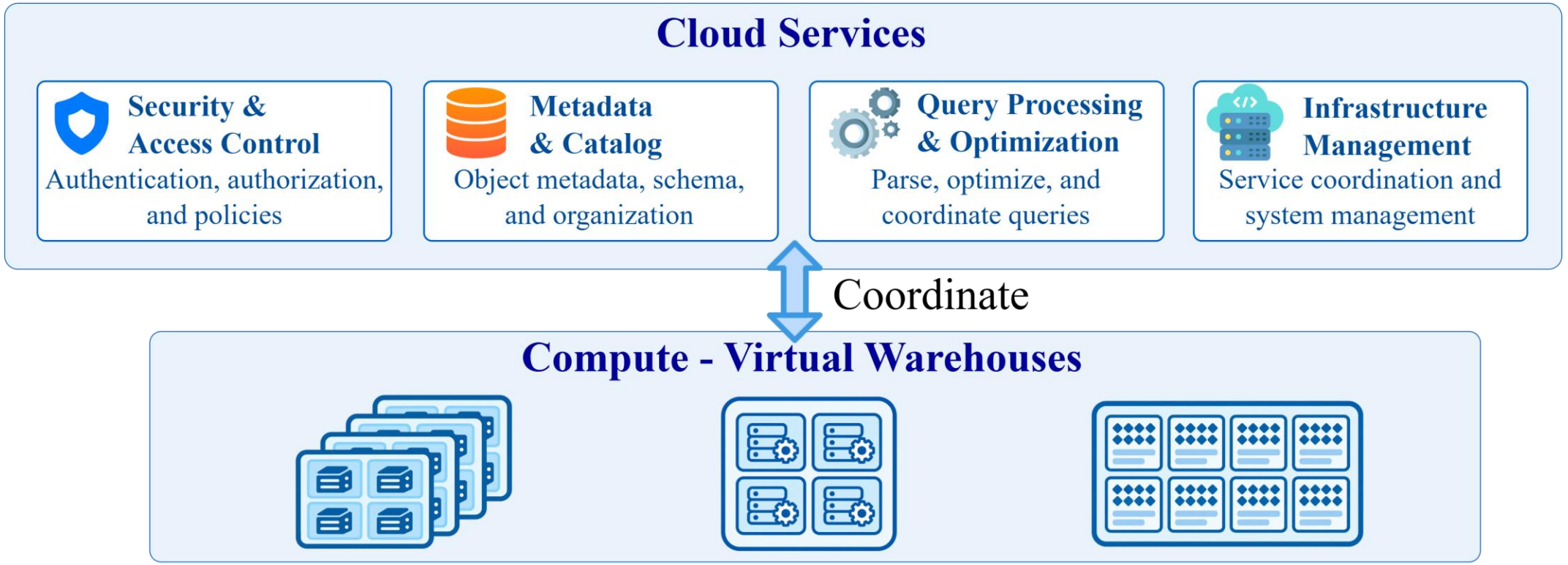


Virtual Warehouses provide independent compute for different workloads while sharing the same underlying data storage.

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Snowflake

❖ Snowflake Architecture: Cloud Services

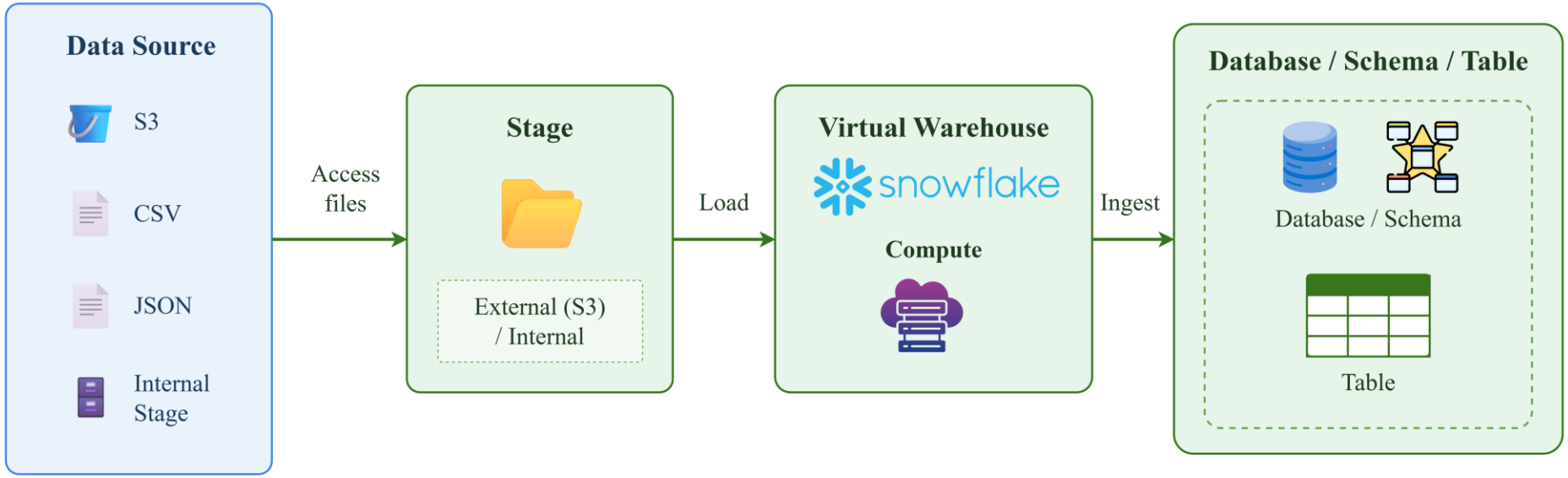


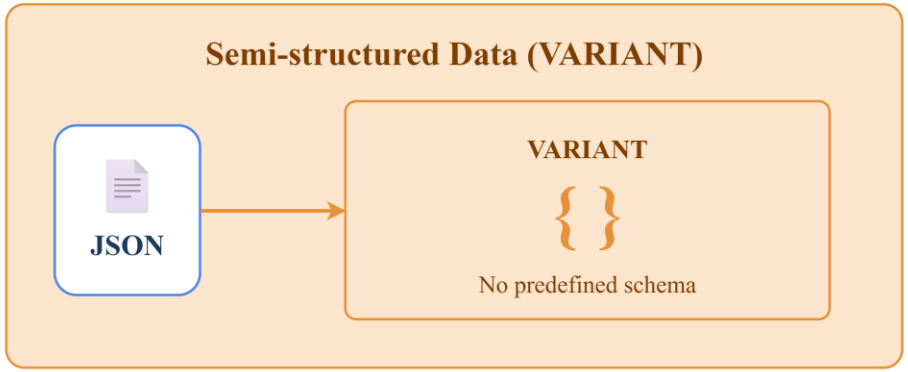
Cloud Services coordinates Snowflake by managing access, metadata, query planning, and platform operations before compute processes the data.

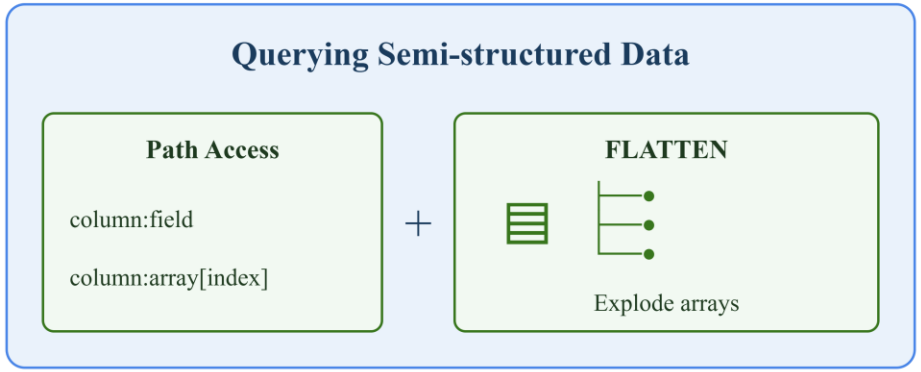
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Snowflake

❖ Stage & Semi-structured Data in Snowflake







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Snowflake

❖ Key Snowflake Concepts

Virtual Warehouse



Independent compute used to run queries and workloads

Micro-partition



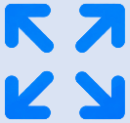
Small storage units that organize table data efficiently

Workload Isolation



Different workloads can use separate warehouses

Independent Scaling



Compute and storage can scale separately

Snowflake separates shared data from independent compute, allowing workloads to scale and run with less interference.

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Snowflake

❖ Snowflake Core & Data Loading

Architecture & Core Objects

Feature	What to remember
Storage	Centralized cloud storage for data
Virtual Warehouse	Independent compute for queries and loading
Cloud Services	Metadata, security, optimization, orchestration
Database / Schema	Organize Snowflake objects
Table / View	Store or expose data
Stage	Location for files used in loading / unloading

Data Ingestion & Staging

Feature	What to remember
COPY INTO	Batch load files into tables
Internal Stage	Files stored in Snowflake-managed staging
External Stage	Files in S3, Azure Blob, or GCS
File Format	Defines CSV / JSON / Parquet parsing
Snowpipe	Continuous automated file ingestion
External Table	Query external files without normal loading

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❖ Transformation & Pipelines

Data Pipelines

Feature	What to remember
Snowpipe	Automated ingestion for arriving files
Streams	Track row-level changes (CDC)
Tasks	Schedule or trigger SQL workflows
Dynamic Tables	Automatically maintain transformed tables
Feature	What to remember
Snowpipe	Automated ingestion for arriving files

SQL & Semi-Structured Data

Feature	What to remember
CTEs	Break complex SQL into readable steps
Window Functions	Row-level analytics across partitions
VARIANT	Store semi-structured data such as JSON
JSON Path	Access nested values like data:user.id
FLATTEN	Expand arrays or objects into rows

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Snowflake

❖ Performance & Data Protection

Performance & Optimization

Feature	What to remember
Micro-partitions	Small columnar storage units that enable pruning
Partition Pruning	Scan only relevant data to improve query speed
Clustering	Improves pruning for large tables with selective queries
Materialized Views	Store precomputed results for frequent complex queries
Result Cache	Reuse previous results when underlying data is unchanged
Query Profile	Inspect execution details and identify bottlenecks

Time Travel & Zero-Copy Clone

Feature	What to remember
Time Travel	Query or restore historical data, including changed or dropped data
Retention Period	Typically 1–90 days depending on edition and settings
Zero-Copy Clone	Instant metadata-only copy without duplicating underlying data
Recovery	Useful for restoring data after mistakes
Dev/Test Sandbox	Create safe environments for testing and experimentation
Versioning	Useful for isolated development and snapshot-like workflows

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Snowflake

❖ Governance & Python/ML

Access Control & Governance

Feature	What to remember
Roles & Privileges	Grant object-level access using RBAC
Row Access Policies	Restrict which rows users can see
Masking Policies	Hide or obfuscate sensitive column values
Network Policies	Restrict login access by IP or network rules
SSO/OAuth/SCIM	Integrate enterprise identity and user provisioning

Snowpark & ML

Feature	What to remember
Snowpark	Use Python, Scala, or Java to work with Snowflake data
Snowpark DataFrame	DataFrame API for transformations close to the data
Python UDFs	Run custom Python logic inside Snowflake
Snowpark ML	Support ML preprocessing and model workflows
Model Registry	Track model versions and manage inference workflows
Feature	What to remember


QUIZ
TIME

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Practice Project

Vietnam Air Quality

❖ Overview

Business Requirement

- Highest pollution city?
- AQI trend (hourly)
- Dry vs. rainy season
- Highest pollution station?

Multi-source Data Source

WAQI API

- JSON (nested)
- Real-time
- 3 times/day
- Slug (e.g. ha-noi)

Different format

Kaggle CSV

- Table (flat)
- Historical 2021
- Readable name (e.g. Ha Noi)

Integration Challenges

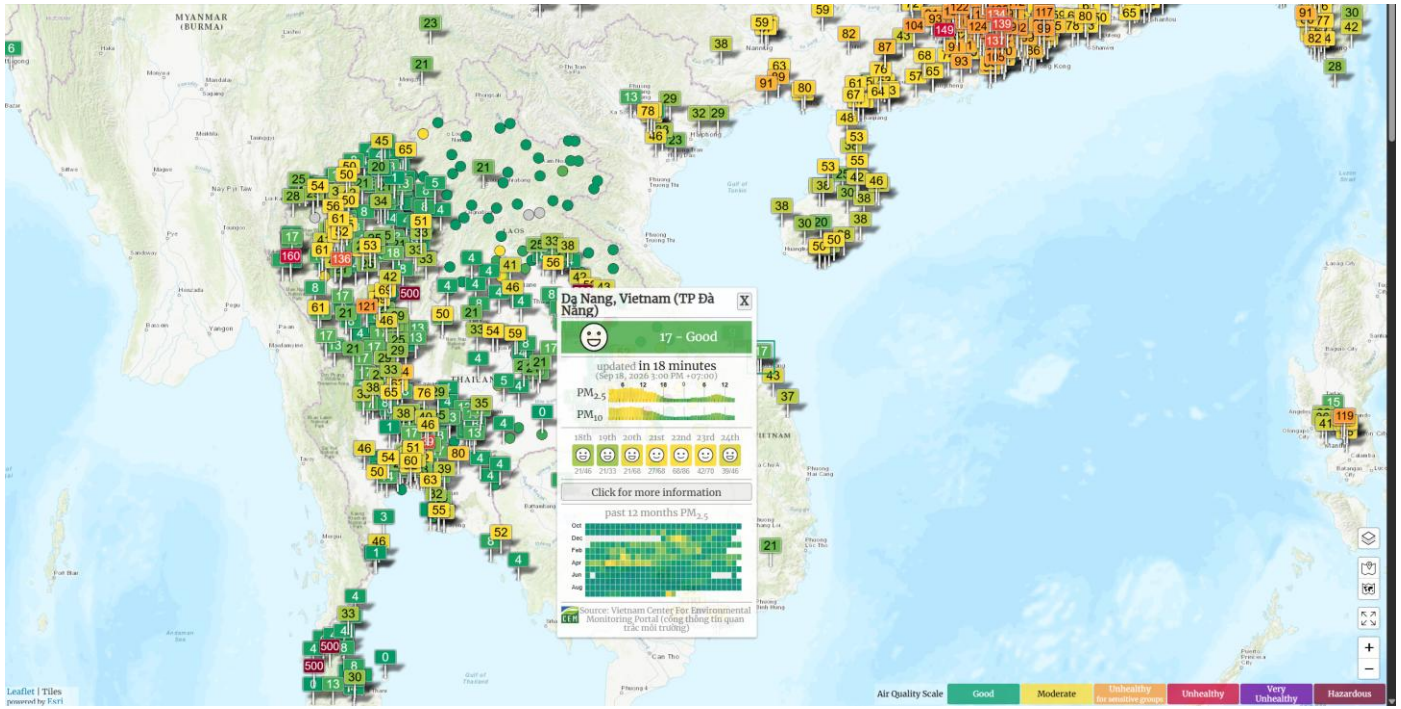
- ⚠ City name mismatch
- ⚠ Parse JSON repeatedly
- ⚠ Higher compute cost

The project use only Snowflake-native features: VARIANT + FLATTEN, Streams (change data capture), Tasks (scheduler), SQL stored procedures, Time Travel, Zero-Copy Clone, automatic micro-partitioning, and a resource monitor for cost control.

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Vietnam Air Quality

❖ Dataset



https://aqicn.org/api/

https://waqi.info/

API_TEMPLATE = "https://api.waqi.info/feed/{city}/?token={token}"

```python
def _fetch_city(city: str, token: str) -> dict | None:
    url = API_TEMPLATE.format(city=city, token=token)
    resp = requests.get(url, timeout=30)
    resp.raise_for_status()
    payload = resp.json()

    if payload.get("status") != "ok":
        print(f"[waqi] {city}: status={payload.get('status')} data={payload.get('data')}")
        return None

    # Embed the queried city so Bronze preserves the request context alongside
    # the API payload — exactly what the AWS partition key `queried_city` did.
    data = payload["data"]
    data["_queried_city"] = city
    return data
```

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Vietnam Air Quality

❖ Dataset Sample


```
{
  "status": "ok",
  "data": {
    "aqi": 62.3,
    "idx": 10241,
    "dominentpol": "pm25",
    "time": {
      "iso": "2026-01-01T08:00:00Z"
    },
    "city": {
      "name": "Da Nang City Center",
      "url":
"https://aqicn.org/station/10241",
      "geo": [
        16.07,
        108.22
      ]
    },
    "iaqi": {
      "pm25": {
        "v": 56.5
      },
      "pm10": {
        "v": 92.6
      },
      "co": {
        "v": 722.0
      },
      "no2": {
        "v": 9.8
      },
      "o3": {
        "v": 57.0
      },
      "so2": {
        "v": 6.6
      },
      "h": {
        "v": 67.0
      },
      "t": {
        "v": 21.0
      },
      "p": {
        "v": 1017.3
      },
      "w": {
        "v": 0.7
      }
    },
    "_queried_city": "da-nang"
  }
}
```


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	station_id	aqi_index	location	station_nan	url	dominant_p	co	dew	humidity	no2	o3	pressure	pm10	pm25	so2	temperatur	wind	data_time_s	data_time_t	status	alert_level
2	10221	15	Ha Noi	US Embassy	https://aqic	pm25	322	25.9	69	39.6	53.3	1001	23.1	13.9	10.5	30.9	2	########	Asia/Ho_Ch	ok	good
3	10221	15	Ha Noi	US Embassy	https://aqic	pm25	714.9	21.1	73.7	30	59.4	1018.6	25.5	15.5	4.6	26.1	0.8	########	Asia/Ho_Ch	ok	good
4	10221	15	Ha Noi	US Embassy	https://aqic	pm25	340.6	22.7	64	23.8	47.5	1005.5	22.7	14.6	5.6	27.7	2.3	########	Asia/Ho_Ch	ok	good
5	10221	20.4	Ha Noi	US Embassy	https://aqic	pm25	365.6	22.6	74.5	35	51.4	1018.7	34.9	21.2	11.9	27.6	4	########	Asia/Ho_Ch	ok	good
6	10221	16.7	Ha Noi	US Embassy	https://aqic	pm25	615.2	20.3	76.9	44.8	37.7	1012.6	26.1	17.1	4.2	25.3	2.2	########	Asia/Ho_Ch	ok	good
7	10221	19.6	Ha Noi	US Embassy	https://aqic	pm25	887.6	24.9	63.5	8.1	21.2	1016	33.7	20	8.6	29.9	4	########	Asia/Ho_Ch	ok	good
8	10221	27.1	Ha Noi	US Embassy	https://aqic	pm25	682.5	17.3	83.7	8.4	30.1	1010.1	42.6	26.2	5.5	22.3	1.6	########	Asia/Ho_Ch	ok	good
9	10221	21.4	Ha Noi	US Embassy	https://aqic	pm25	734.6	17.7	69.9	30.8	21.5	1013.4	34	22.1	4.6	22.7	3.7	########	Asia/Ho_Ch	ok	good
10	10221	35	Ha Noi	US Embassy	https://aqic	pm25	676.3	18.8	83.5	41	46.8	1014.9	57.3	36	6	23.8	3.4	########	Asia/Ho_Ch	ok	good
11	10221	33.7	Ha Noi	US Embassy	https://aqic	pm25	621.5	18	86.6	15.7	54.3	1019.4	53.7	35.2	10.2	23	1.5	########	Asia/Ho_Ch	ok	good
12	10221	29.7	Ha Noi	US Embassy	https://aqic	pm25	805.4	23.5	85.9	29.1	39.5	1014.1	53.6	32.5	8.4	28.5	1.6	########	Asia/Ho_Ch	ok	good
13	10221	33.8	Ha Noi	US Embassy	https://aqic	pm25	442.6	14.6	69.1	24.7	67.5	1015.7	53.1	33.2	4.9	19.6	3.2	########	Asia/Ho_Ch	ok	good
14	10221	49.1	Ha Noi	US Embassy	https://aqic	pm25	589.6	20.3	82	26.5	29.5	1015.5	68.4	48.5	11.3	25.3	1	########	Asia/Ho_Ch	ok	good
15	10221	47.7	Ha Noi	US Embassy	https://aqic	pm25	706	20.3	81.4	12.9	56.2	1000.1	90.2	50.8	7.1	25.3	3.8	########	Asia/Ho_Ch	ok	good
16	10221	49.7	Ha Noi	US Embassy	https://aqic	pm25	580.1	21.1	69.4	9.7	46	1006.2	72.6	50.6	8.9	26.1	3.6	########	Asia/Ho_Ch	ok	good
17	10221	50	Ha Noi	US Embassy	https://aqic	pm25	491.5	13.2	68.2	24.3	25.9	1012.5	70.8	46.9	4.3	18.2	2.3	########	Asia/Ho_Ch	ok	good
18	10221	47.8	Ha Noi	US Embassy	https://aqic	pm25	384.9	21.6	62.1	34.1	34.2	1006.8	70.7	46.3	5.2	26.6	2.1	########	Asia/Ho_Ch	ok	good
19	10221	51	Ha Noi	US Embassy	https://aqic	pm25	363.6	21.3	77.9	10	41.3	1016.7	84.7	50.7	4.6	26.3	2.9	########	Asia/Ho_Ch	ok	moderate
20	10221	61.5	Ha Noi	US Embassy	https://aqic	pm25	727.7	19.5	82.7	32.8	46.7	1002.6	102.2	59.2	11.8	24.5	3.3	########	Asia/Ho_Ch	ok	moderate
21	10221	64.3	Ha Noi	US Embassy	https://aqic	pm25	710.7	25.1	66.7	34.3	43.3	1012.3	101.1	63.9	5.1	30.1	3.7	########	Asia/Ho_Ch	ok	moderate
22	10221	62.1	Ha Noi	US Embassy	https://aqic	pm25	773.3	17.9	74.3	40.6	42.7	1009.2	107.6	64.6	3.3	22.9	1.1	########	Asia/Ho_Ch	ok	moderate

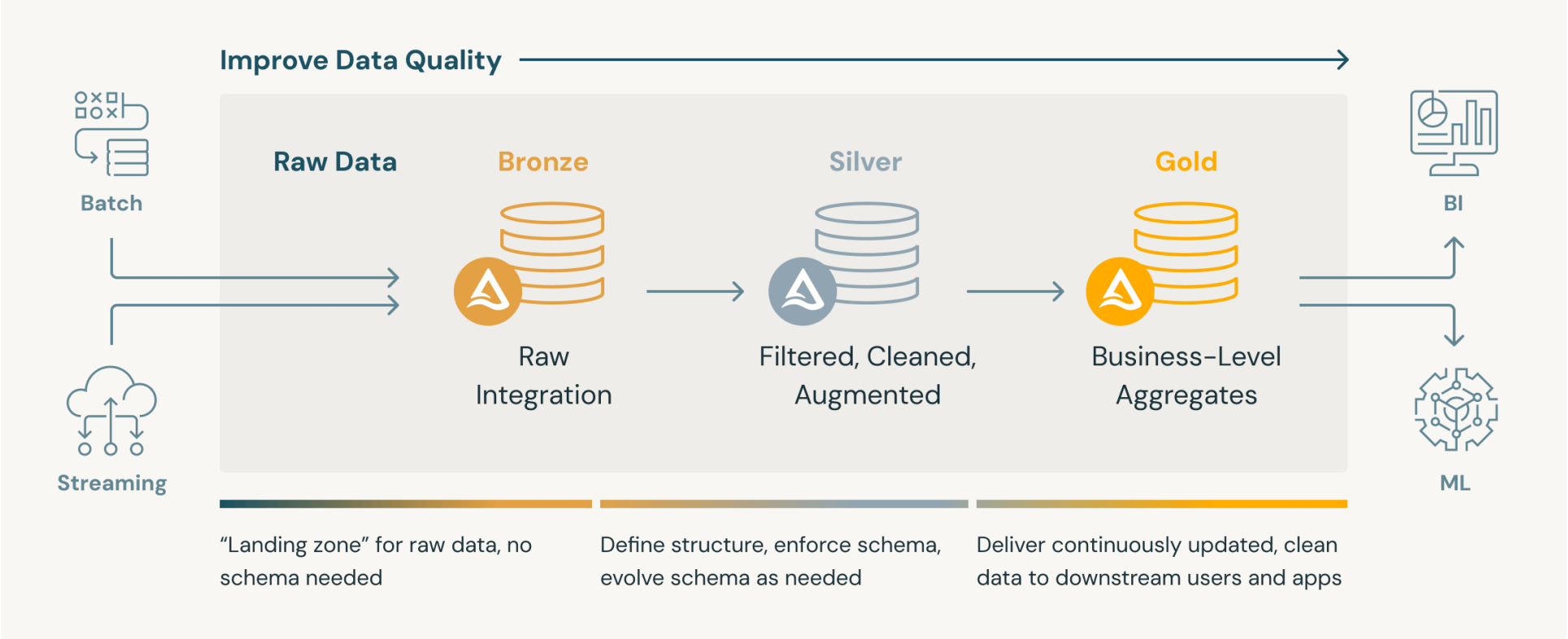
```
CREATE OR REPLACE TABLE
BRONZE.HISTORICAL_CSV (
    station_id            STRING,
    aqi_index             STRING,
    location              STRING,
    station_name          STRING,
    url                   STRING,
    dominant_pollutant    STRING,
    co                    STRING,
    dew                   STRING,
    humidity              STRING,
    no2                   STRING,
    o3                    STRING,
    pressure              STRING,
    pm10                  STRING,
    pm25                  STRING,
    so2                   STRING,
    temperature           STRING,
    wind                  STRING,
    data_time_s           STRING,
    data_time_tz          STRING,
    status                STRING,
    alert_level           STRING,
    ingested_at           TIMESTAMP_NTZ
DEFAULT CURRENT_TIMESTAMP()
)
COMMENT = 'Bronze: raw 2021 Kaggle
historical CSV, types preserved as STRING';
```

```
CREATE OR REPLACE TABLE BRONZE.API_RAW (
    queried_city   STRING    NOT NULL COMMENT 'City we asked the API for',
    raw_payload    VARIANT   NOT NULL COMMENT 'Full, untouched WAQI JSON response',
    ingested_at    TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP() COMMENT 'Load time',
    source_file    STRING COMMENT 'Original file name (auditability)
)
COMMENT = 'Bronze: raw WAQI API responses as VARIANT (ELT-style raw preservation)';
```

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❖ Medallion Architecture

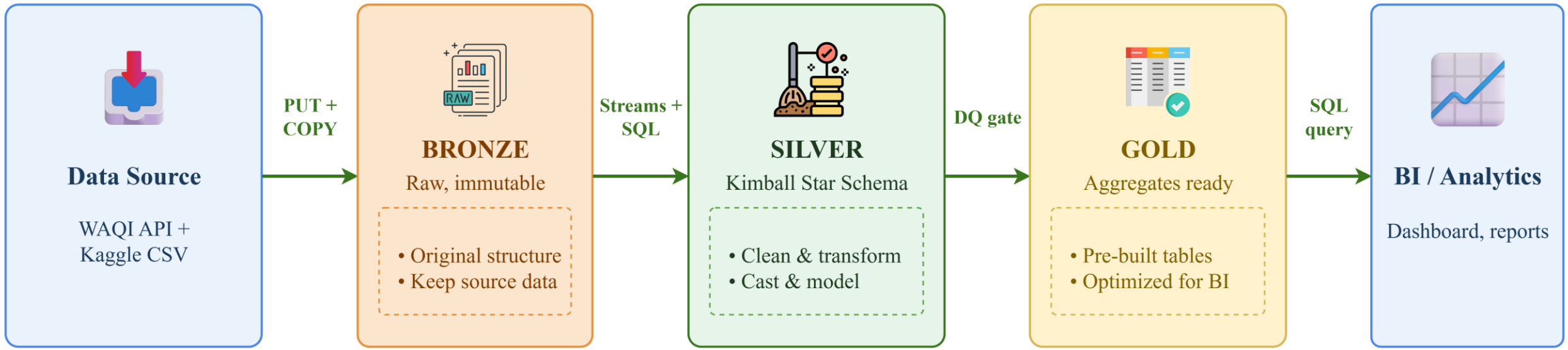


A medallion architecture is a data design pattern used to logically organize data in a lakehouse, with the goal of incrementally and progressively improving the structure and quality of data as it flows through each layer of the architecture. (*Databricks*)

References

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❖ Medallion Architecture



Medallion in our project

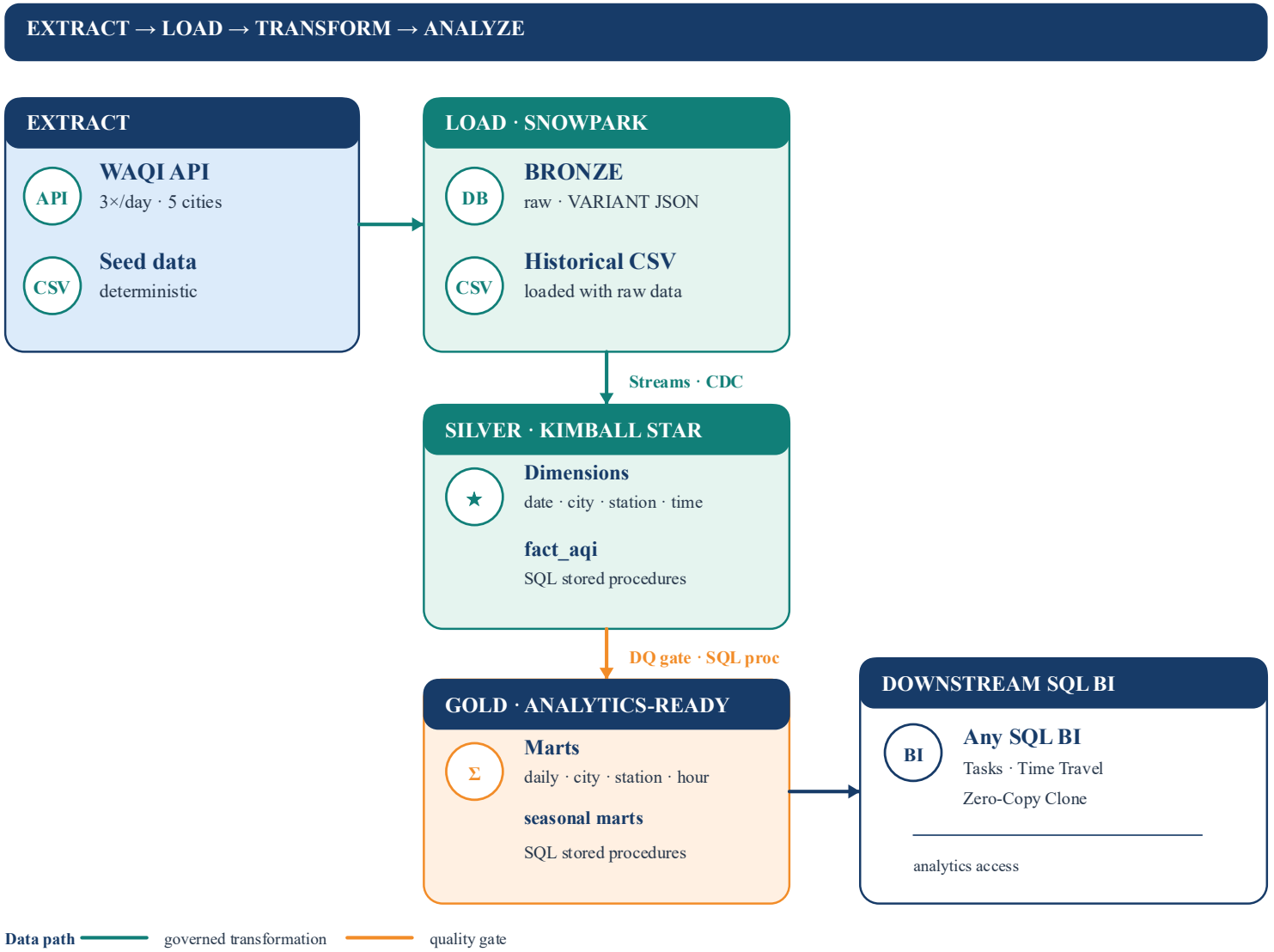
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❖ Detail Architecture

SNOWPARK

[Snowpark](#) is a developer framework by Snowflake that allows you to write and run code in languages like Python, Java, and Scala directly inside your cloud data platform

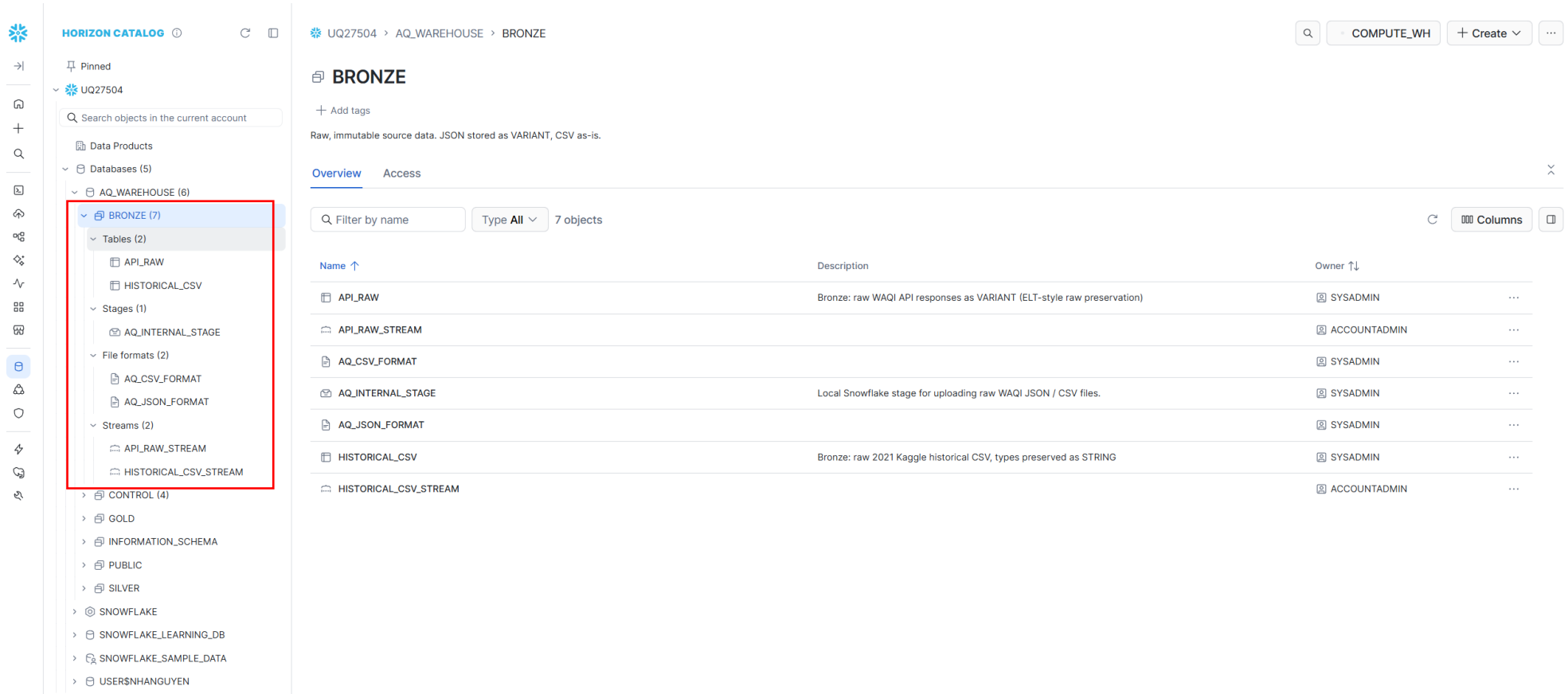


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❖ Broze Layer

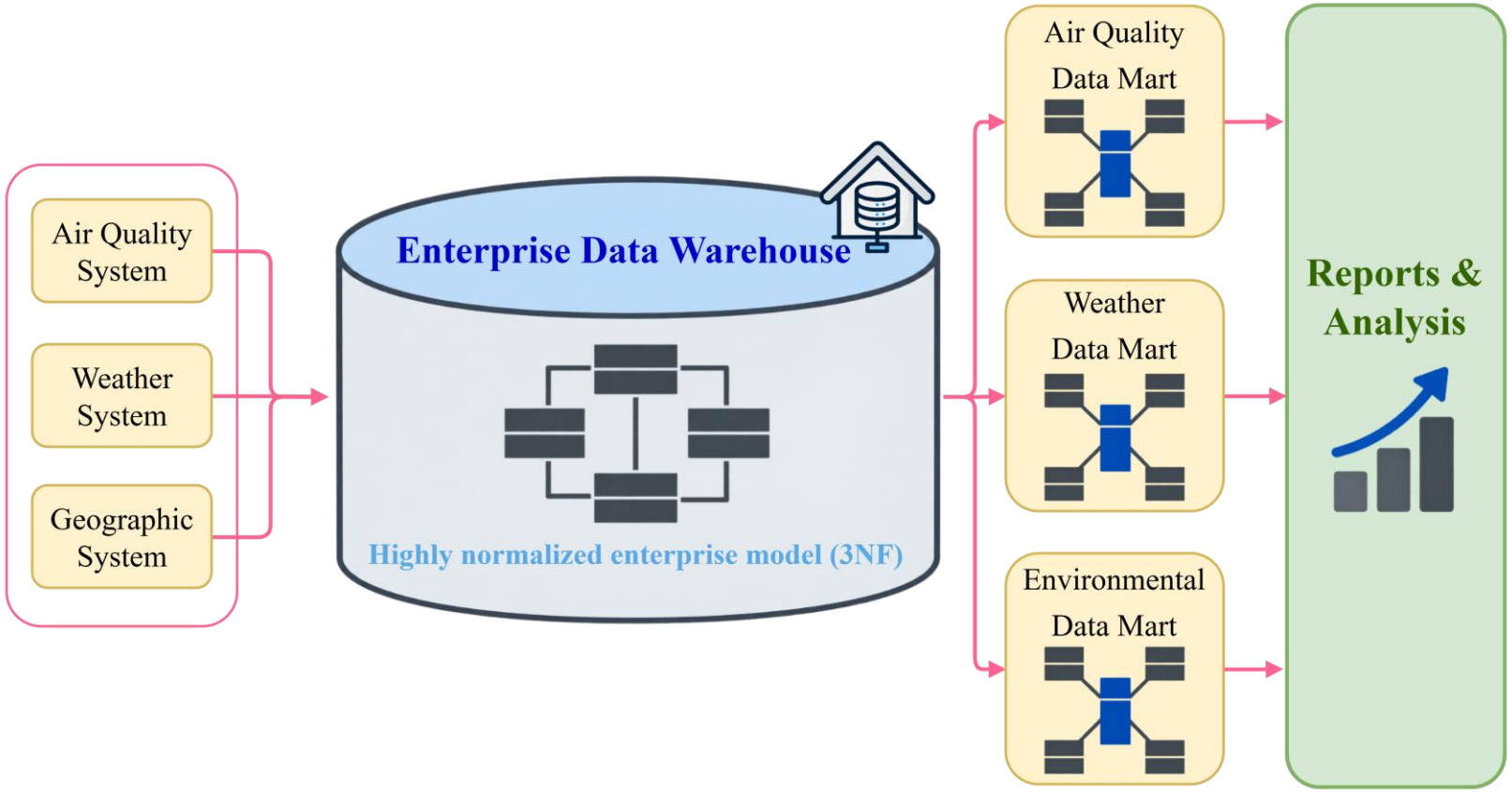
Table	Grain	Notes
`BRONZE.API_RAW`	one WAQI API response	full JSON kept in a `VARIANT` column
`BRONZE.HISTORICAL_CSV`	one 2021 CSV record	all columns kept as `STRING`



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Database to Data Warehouse

❖ Silver Layer



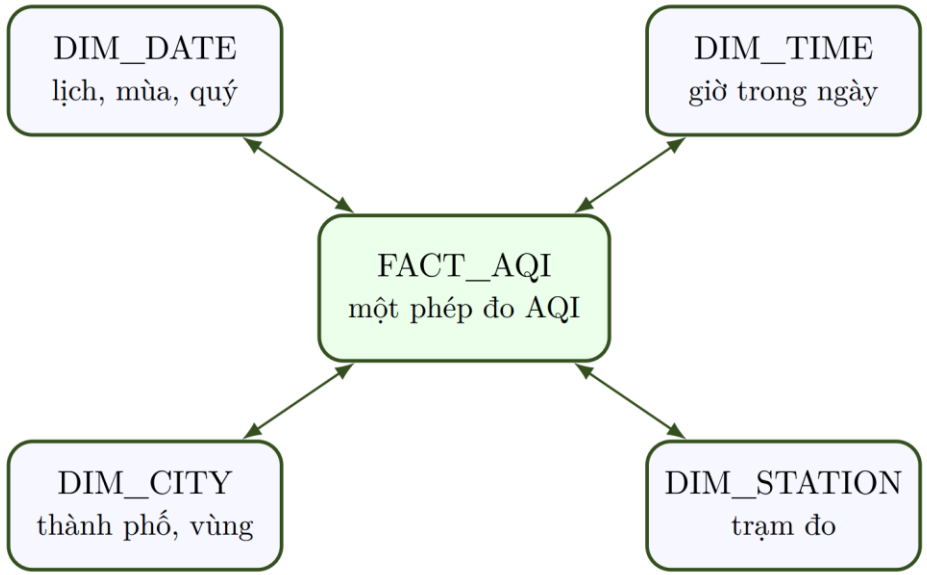
Build dimensional data marts around business processes first, then integrate them into an enterprise-wide warehouse.

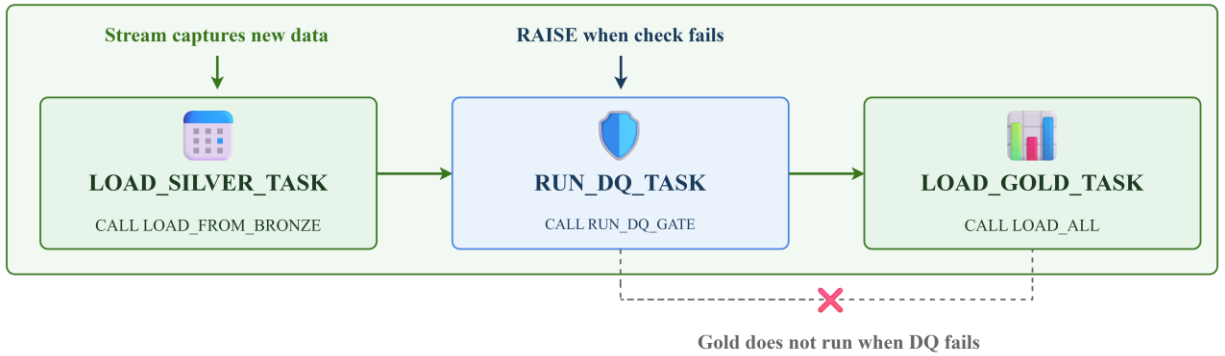
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❖ Silver

Table	Grain	Notes
SILVER.DIM_DATE	one calendar date	conformed calendar dimension (date_key = YYYYMMDD), season/quarter/weekday
SILVER.DIM_TIME	one hour of day	conformed time-of-day dimension (time_key = HHMM), morning/afternoon/evening
SILVER.DIM_CITY	one WAQI city slug	conformed city dimension with name/region/center coords
SILVER.DIM_STATION	one station per city	conformed station dimension (surrogate key + natural waqi_idx)
SILVER.FACT_AQI	one measurement at one station/time	references the four dims via surrogate keys



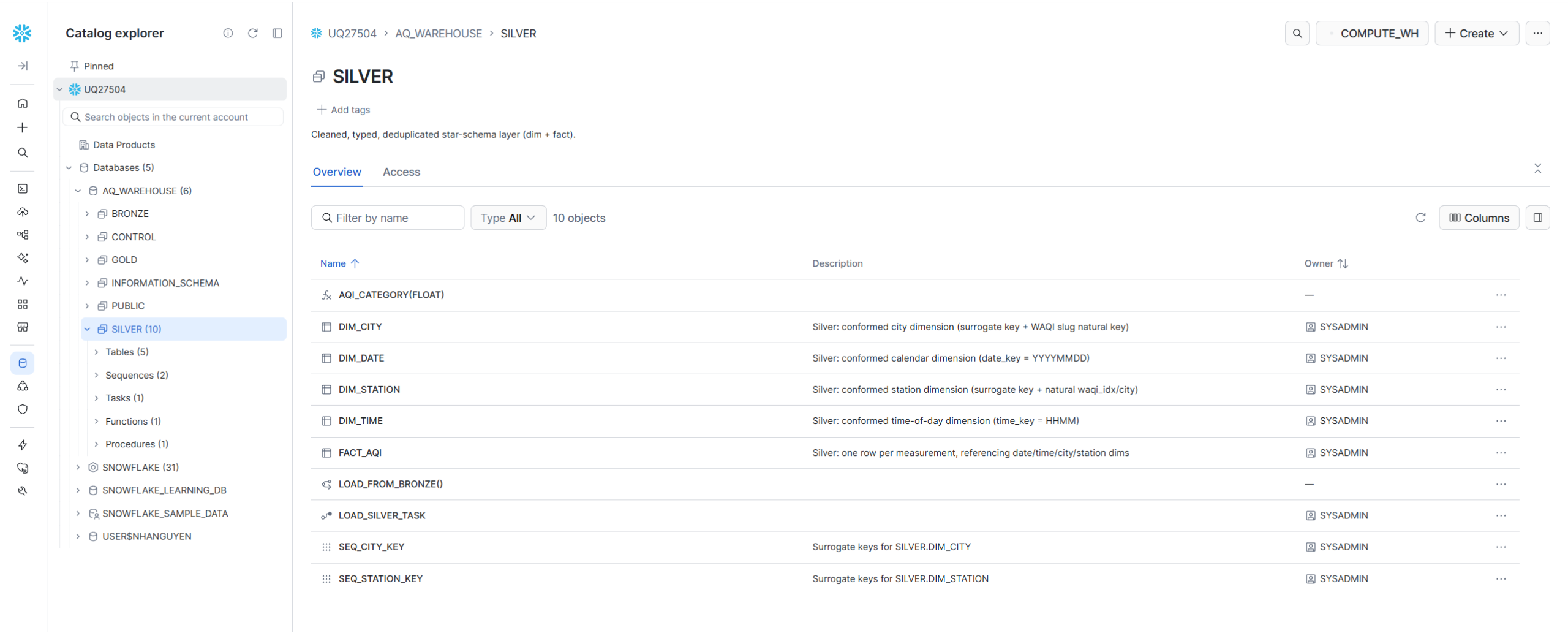


Data Quality Gate

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Vietnam Air Quality

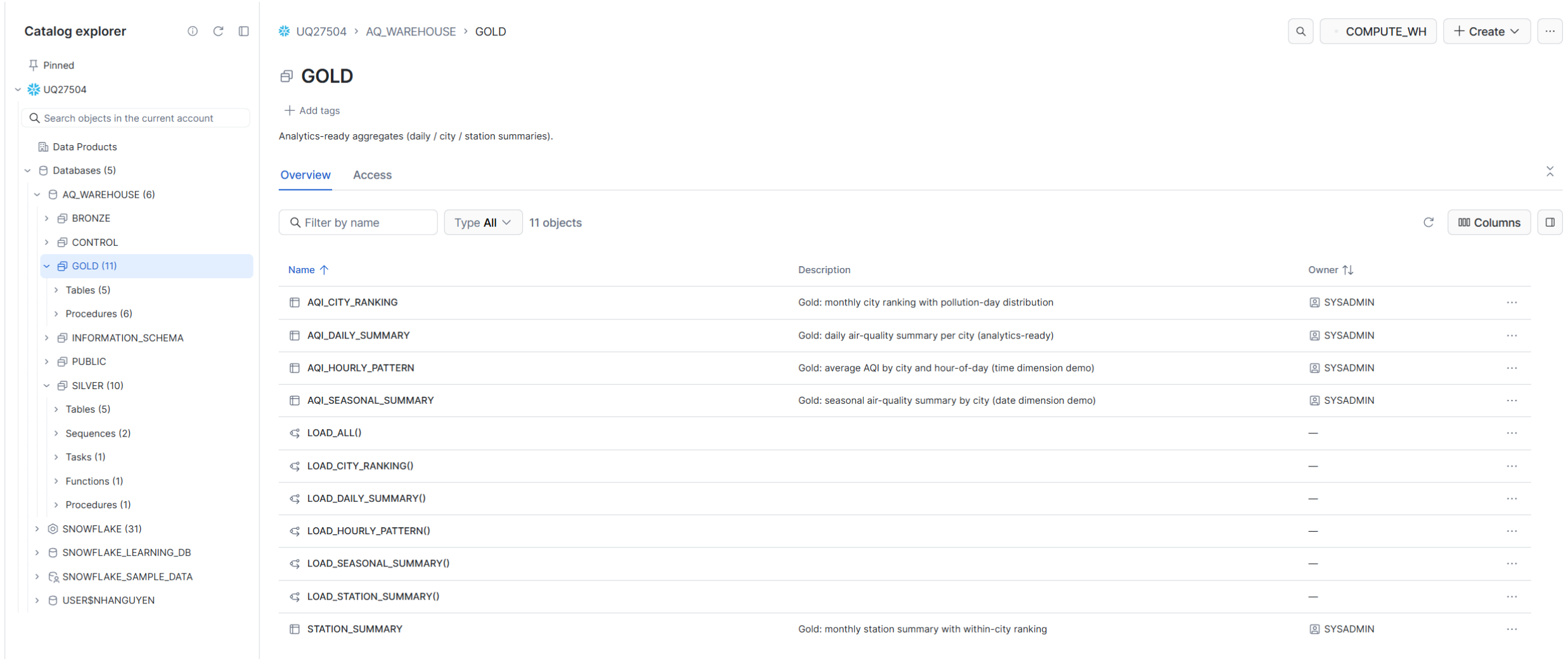
❖ Silver Layer



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❖ Gold Layer



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❖ Result

AQI_CITY_RANKING

+ Add tags

Gold: monthly city ranking with pollution-day distribution

Overview | Preview | Access | Copy history

10 rows | 260ms

	QUERIED_CITY	CITY_NAME	REGION	YEAR	MONTH	AVG_AQI	MAX_AQI	MIN_AQI	AVG_PM25	AVG_PM10	RECORD_COUNT	DAYS_GOOD	DAYS_MODERATE	DAYS_SENSITIVI
	cao-bang 20.0% da-nang 20.0% +3 more	Cao Ba... 20.0% Da Nang 20.0% +3 more	No... 40.0% Ce... 20.0% +2 more	20... 20...	1 1	35.647... 140.85...	66 170	15 111.8	35.738... 140.22...	57.487...223.26...	30 30	0 22	0 27	0
1	cao-bang	Cao Bang	North	2026	1	140.856666667	170	111.8	140.228888889	223.261111111	30	0	0	
2	cao-bang	Cao Bang	North	2021	1	123.379444444	157.2	94.8	123.666111111	197.726111111	30	0	0	
3	gia-lai	Gia Lai	Central Highlar	2026	1	116.238888889	149.2	88.3	116.152222222	186.558888889	30	0	5	
4	gia-lai	Gia Lai	Central Highlar	2021	1	101.330555556	135.2	73.2	101.783888889	161.631111111	30	0	13	
5	da-nang	Da Nang	Central	2026	1	90.533333333	121.7	60.8	90.412222222	145.921111111	30	0	21	
6	da-nang	Da Nang	Central	2021	1	79.523888889	112	49.5	80.022777778	128.581111111	30	0	22	
7	ho-chi-minh-city	Ho Chi Minh City	South	2026	1	65.891111111	96.8	36.6	65.662222222	105.362222222	30	3	27	
8	ho-chi-minh-city	Ho Chi Minh City	South	2021	1	57.282962963	89.5	27.4	57.44037037	92.48037037	30	10	20	
9	ha-noi	Ha Noi	North	2026	1	40.966666667	74.1	15	40.42	64.918888889	30	19	11	
10	ha-noi	Ha Noi	North	2021	1	35.647777778	66	15	35.738148148	57.487407407	30	22	8	

Monthly city ranking with pollution-day distribution

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❖ Result

AQI_SEASONAL_SUMMARY

Add tags

Gold: seasonal air-quality summary by city (date dimension demo)

Overview | Preview | Access | Copy history

10 rows | 241ms

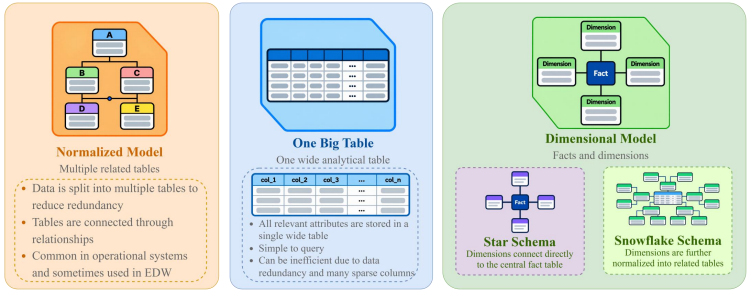
	QUERIED_CITY	CITY_NAME	REGION	YEAR	SEASON	AVG_AQI	MAX_AQI	MIN_AQI	AVG_PM25	AVG_PM10	RECORD_COUNT	DOMINANT_POLLUTANT	AGGREGATED_AT
	cao-bang 20.0% da-nang 20.0% +3 more	Cao Ba... 20.0% Da Nang 20.0% +3 more	North 40.0% Central 20.0% +2 more	20... 20...	Dry 100.0%	35.647... 140.85...	66 170	15 111.8	35.738... 140.22...	57.487...223.26...	90 270	pm25 100.0%	9/18/2026 9/18/20
1	gia-lai	Gia Lai	Central Highlands	2021	Dry	101.330555556	135.2	73.2	101.783888889	161.631111111	180	pm25	2026-09-18 17:54:47.1
2	da-nang	Da Nang	Central	2026	Dry	90.533333333	121.7	60.8	90.412222222	145.921111111	90	pm25	2026-09-18 17:54:47.1
3	ha-noi	Ha Noi	North	2021	Dry	35.647777778	66	15	35.738148148	57.487407407	270	pm25	2026-09-18 17:54:47.1
4	da-nang	Da Nang	Central	2021	Dry	79.523888889	112	49.5	80.022777778	128.581111111	180	pm25	2026-09-18 17:54:47.1
5	ha-noi	Ha Noi	North	2026	Dry	40.966666667	74.1	15	40.42	64.918888889	90	pm25	2026-09-18 17:54:47.1
6	ho-chi-minh-city	Ho Chi Minh City	South	2026	Dry	65.891111111	96.8	36.6	65.662222222	105.362222222	90	pm25	2026-09-18 17:54:47.1
7	ho-chi-minh-city	Ho Chi Minh City	South	2021	Dry	57.282962963	89.5	27.4	57.44037037	92.48037037	270	pm25	2026-09-18 17:54:47.1
8	cao-bang	Cao Bang	North	2026	Dry	140.856666667	170	111.8	140.228888889	223.261111111	90	pm25	2026-09-18 17:54:47.1
9	cao-bang	Cao Bang	North	2021	Dry	123.379444444	157.2	94.8	123.666111111	197.726111111	180	pm25	2026-09-18 17:54:47.1
10	gia-lai	Gia Lai	Central Highlands	2026	Dry	116.238888889	149.2	88.3	116.152222222	186.558888889	90	pm25	2026-09-18 17:54:47.1

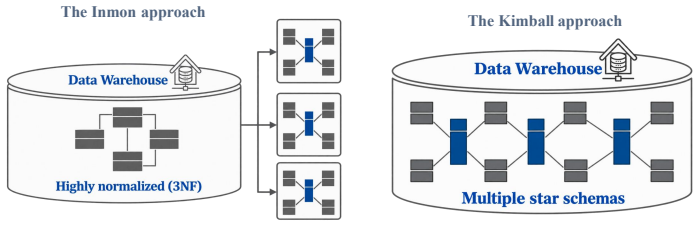
Seasonal air-quality summary by city (date dimension)

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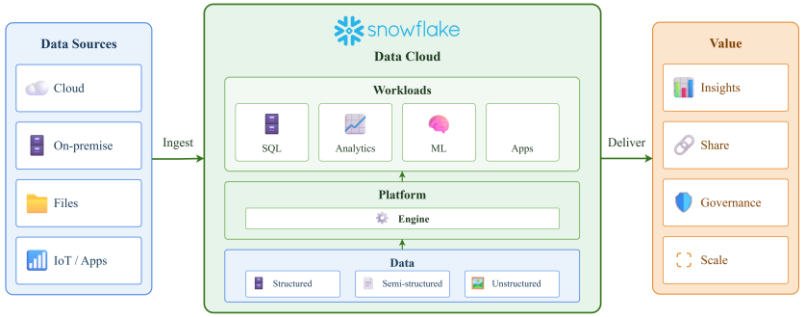
Lesson: Summary

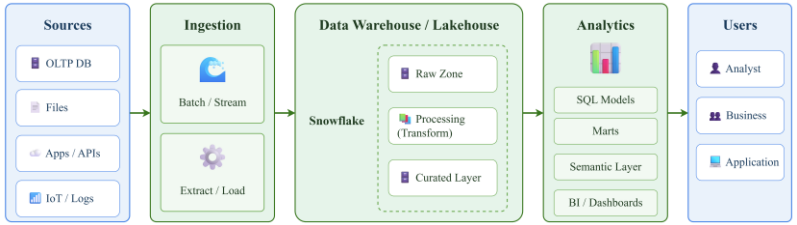
Data Warehouse



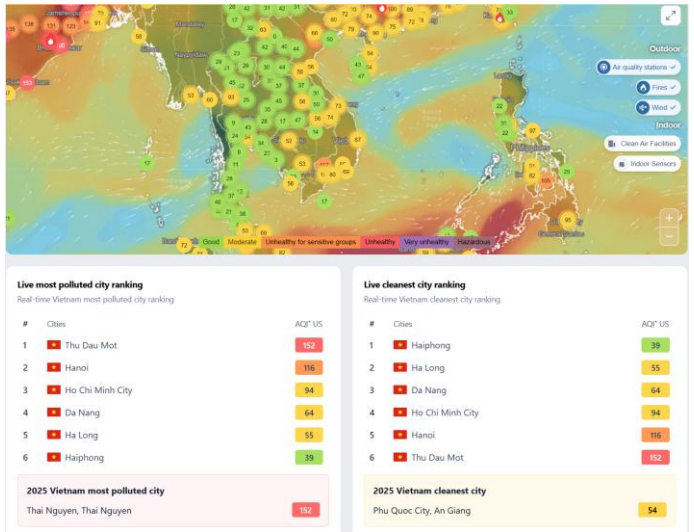


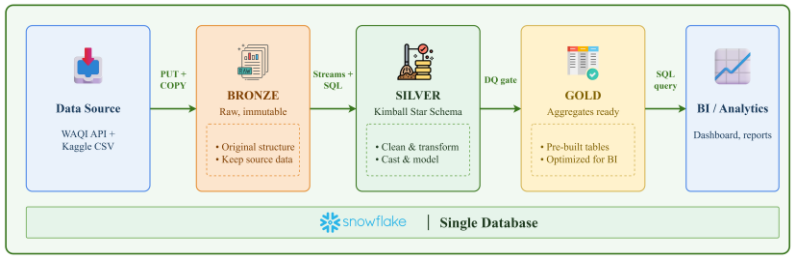
Snowflake





Vietnam Air Quality





Review

- Introduction
- From Database to Data Warehouse
- Snowflake
- Project: Vietnam Air Quality Data Pipeline
- Q&A

Quiz&Feedbacks


Thank you!